

Concord is Insatiable Agree¹

Heterogeneous Case and Phi Patterns in Slavic and Finnic

Abstract: This paper argues that nominal concord does not require dedicated feature-percolation mechanisms. Building on the interaction/satisfaction model of Agree and on the *horizons*-based theory of locality, it proposes that concord arises when an insatiable probe successively interacts with all accessible feature-bearing constituents in an extended nominal projection, transmitting case and phi features. The analysis naturally derives standard concord patterns and a typological asymmetry between case concord and phi concord. It is then applied to heterogeneous case patterns in Slavic numeral–noun constructions, where case stacking and a markedness-based rule for morphological realization account for the contrast between direct and oblique cases. The proposed account extends to Russian paucal constructions under both the “paucal case” hypothesis and the “paucal number” approach defended in the paper. Finally, Estonian pseudopartitives are shown to be compatible with the proposed approach to concord once the Estonian case hierarchy is independently recognized to be partially ordered, with genitive and partitive incomparable in markedness. The resulting account unifies a range of apparently highly idiosyncratic Slavic and Finnic concord phenomena using independently motivated theoretical mechanisms and refining some of these mechanisms in the process.

Keywords: nominal concord, Agree, case stacking, Slavic numerals, Russian paucals, Estonian pseudopartitives

1 Introduction

Minimalism strives to explain the richness of syntactic phenomena with a minimal number of theoretical constructs. Two operations are widely assumed to be irreducible, although their exact nature and possible variants are still a matter of lively debate: Merge (for building structures) and Agree (for establishing dependencies between constituents). The need for some kind of locality theory is also widely accepted. Other mechanisms are treated as suspect, and the goal of reducing them to Merge, Agree, and/or locality is generally deemed worth pursuing.

One area that has resisted such reduction is concord, understood as agreement within nominal constituents, typically in case, number, and gender. As emphasized in Baker 2008, concord is similar to other kinds of agreement, usually analyzed via Agree, and yet it is commonly assumed that other, more specialized feature percolation mechanisms are needed in order to successfully account for nominal concord (e.g., Norris 2014, 2018, Bayırlı 2017, Grabovac 2022).

The first aim of this paper is to show that recent approaches to Agree (Preminger 2014, Deal 2015, 2024, 2025, 2026, Clem 2019, 2023, Clem and Deal 2026; briefly introduced in §2 below) predict nominal concord almost out of the box; §3. The second aim is to demonstrate that – given the ϕ -feature-stacking aspect of Agree (e.g., Deal 2015, Hammerly 2020, Clem 2023, Grishin 2024, Clem and Deal 2026) and the well-known phenomenon of case stacking (e.g., Richards 2013, Pesetsky 2013, Levin 2017, Chen 2018 and references therein) – this account naturally extends to the puzzling heterogeneous vs. homogeneous case patterns in Slavic numeral–noun constructions (e.g., Babby 1985, 1987, Franks 1994, 1995); §4. In particular, this approach makes it possible to explain the truly mind-boggling pattern of ϕ -features in Russian paucal numeral constructions (e.g., Mel’čuk 1985, Pesetsky 2013); §5. The resulting analysis is extended to the Finnic – especially Estonian – data that motivated a prominent specialized approach to concord (Norris 2014, 2018); §6. The conclusion is that concord is simply insatiable Agree and does not necessitate any special feature percolation principles; §7.

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2 Agree

Some recent approaches to Agree reject the idea that the aim of Agree is to value uninterpretable features on pain of derivation failure (Chomsky 2000, 2001). On these approaches, Agree failure does not lead to ungrammaticality, but results in “default agreement” (Preminger 2014). Moreover, given the richness and diversity of agreement phenomena across languages, Agree is taken to be a relatively robust operation, specifiable both in terms of interaction (which features of a goal the probe interacts with) and satisfaction (which features of a goal, if any, make the probe stop looking for further goals) (Deal 2015 *et seq.*). Importantly, probing multiple goals may result in a number of feature bundles on the probe (as well as on the goal), with the effect that the collection of features on a constituent has a more articulate structure than just an unordered set of features (e.g., Deal 2015, Hammerly 2020, Clem 2023, Grishin 2024).

These aspects of the relevant approaches to Agree may be illustrated with Clem and Deal’s (2026) analysis of ergative in Shawi (spoken in Peru). In this language, ergative only appears when the subject is at least as high as the object on the person hierarchy ($1 > 2 > 3$, where ‘>’ means ‘higher than’), and may then be accompanied by an explicit realization of the object’s features on the subject. This is shown in (1)–(2), originally from Bourdeau 2015 (Clem and Deal 2026: 272, (6a)–(7a)):²

- | | |
|---|--|
| <p>(1) I’wara ka-ri-nke pera-ra(-w)-nke.
 yesterday 1-ERG-2.o call-IND-1-2.o
 ‘I called you yesterday.’ (CBT)</p> | <p>(2) I’wara kema(*-ri) nu’wi-r-an-ku.
 yesterday 2(*-ERG) tell.off-IND-2-1.o
 ‘You told me off yesterday.’ (CBT)</p> |
|---|--|

In (1), a 1st person subject acts on a 2nd person (i.e., lower) object. As a result, the subject is marked with ERG (-ri) and, optionally, with ϕ features of the object (-nke). In (2), a 2nd person subject acts on a 1st person (i.e., higher) object, and the subject cannot be marked with either.

Clem and Deal (2026) – assuming cyclic Agree (Béjar and Řezáč 2009) and the interaction/satisfaction framework (Deal 2015, 2024) – analyze (1)–(2) by postulating that a probe on v , which is situated between the subject and the object, is specified as [INT: ϕ , SAT:SPEAKER]: it copies all ϕ features from any element it interacts with, but this interaction stops upon encountering SPEAKER, i.e., 1st person. This is what happens in (2); the probe flags the object with its features (only v , at this stage) and simultaneously copies over (agrees with) the object’s ϕ features. As the object is 1st person, the SAT:SPEAKER condition on the probe is met and probing stops here. By contrast, in (1) the object is not 1st person, so probing does not stop here; a projection of the same v probe additionally flags the subject with its features – which now crucially include the object’s ϕ features in addition to the categorial v feature – and copies the subject’s ϕ features.

The resulting features of the key items in (1)–(2) are schematically shown in (3)–(4), where ϕ_o refers to the object’s ϕ features, and ϕ_s to the subject’s features.³

- | | |
|--|--|
| <p>(3) transitive 1→2 in Shawi (as in (1)):
 S (ERG): [D, ϕ_s] [v, ϕ_o]
 v: [v] [ϕ_o] [ϕ_s]
 O (ABS): [D, ϕ_o] [v]</p> | <p>(4) transitive 2→1 in Shawi (as in (2)):
 S (ABS): [D, ϕ_s]
 v: [v] [ϕ_o]
 O (ABS): [D, ϕ_o] [v]</p> |
|--|--|

The first bundle on each item represents its native features, while subsequent bundles are the result of consecutive probe–goal interactions. The bundle that is crucial in this analysis is the boxed [v , ϕ_o] on the subject in (3), containing ϕ_o – the ϕ features of the object. It is the presence of two sets of ϕ features – ϕ_s and ϕ_o – that is expounded in the context of D as the ergative suffix -ri.

Clem and Deal (2026: §5) extend this general approach to accusative languages. There, the relevant probe is assumed to be above the subject, on T. Hence, this probe first interacts with the subject and then flags the object with

²All examples are annotated with 3-letter ISO 639-3 codes of respective languages: CBT = Shawi, DAN = Danish, DEU = German, EST = Estonian, FIN = Finnish, GOT = Gothic, KHA = Khasi, MIQ = Miskitu, POL = Polish, RUS = Russian. Glossing conventions and abbreviations generally follow the Leipzig Glossing Rules, with o in (1)–(2) additionally indicating agreement with object. Additional abbreviations (e.g., of Estonian cases in §6.1) may be found in https://en.wikipedia.org/wiki/List_of_glossing_abbreviations#Grammatical_abbreviations.

³In (3)–(5), nominal constituents are assumed to be of category D, while S and O indicate subject and object.

the subject's ϕ features; see (5). On this view, ACC is keyed to the presence of two ϕ feature bundles – ϕ_o and ϕ_s – on a DP object.

(5) transitive in accusative languages:

T: [T] [ϕ_s] [ϕ_o]

S (NOM): [D, ϕ_s] [T]

O (ACC): [D, ϕ_o] [T, ϕ_s]

While the probe that Clem and Deal (2026) postulate in their analysis of Shawi is [INT: ϕ , SAT:SPEAKER], they note that the more common languages, in which the case of one argument does not depend on the person of another, require a probe with no satisfaction constraints: [INT: ϕ , SAT:–]. Such a probe is insatiable: it interacts with all accessible ϕ -bearing elements (e.g., Clem 2023, Deal 2024). In the next section, we will see that such insatiable probes may be responsible for concord.

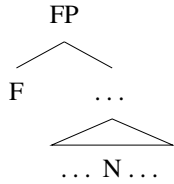
3 Concord

3.1 Extended Nominal Projections

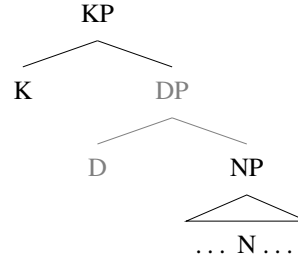
It turns out that the view of Agree presented in the previous section predicts the possibility of concord on almost any approach to the structure of nominal phrases.⁴ The only crucial assumption that needs to be made is the prevalent assumption that maximal nominal projections are headed by a functional head such as D or K (on which more below), i.e., that each maximal nominal projection has the structure in (6), where F is a nominal functional head capable of hosting the case feature.

Maximal extended nominal projection:

(6) minimal assumption, schematically:



(7) assumed here (to be extended):



The dominant view since Abney 1987 is that F is D(eterminer). However, it is also frequently noted that another functional level is needed, corresponding to morphological case, i.e., that F is K(ase) (e.g., Lamontagne and Travis 1987, Bittner 1994, Bittner and Hale 1996a,b, Levin 2015; cf. Caha 2009 *et seq.*). One argument for this assumption is that some languages seem to overtly realize both D and K, in accordance with their general headedness pattern, as in (8) (Khasi – a head-initial Austroasiatic language spoken in India) and (9) (Miskitu – a head-final Misumalpan language spoken in Nicaragua), both from Bittner 1994: 4.

(8) Ka la yo”ii [ya ’u khlaa].
she PST see ACC the tiger
‘She saw the tiger.’ (KHA)

(9) [Waitna ba] [sula ba ra] kaik-an.
man the deer the ACC see-PST.3SG
‘The man saw the deer.’ (MIQ)

The minimal structure of nominal projections in such examples is (7), with all heads overtly realized. Hence – even if, for the purposes of this paper, F in (6) could be any functional head capable of hosting case features – we will follow the increasingly common view that maximal nominal projections are indeed KPs, as in (7).⁵ However, we will not prejudice the NP/DP issue, i.e., we will remain agnostic about (the universality of) the DP nominal projection.

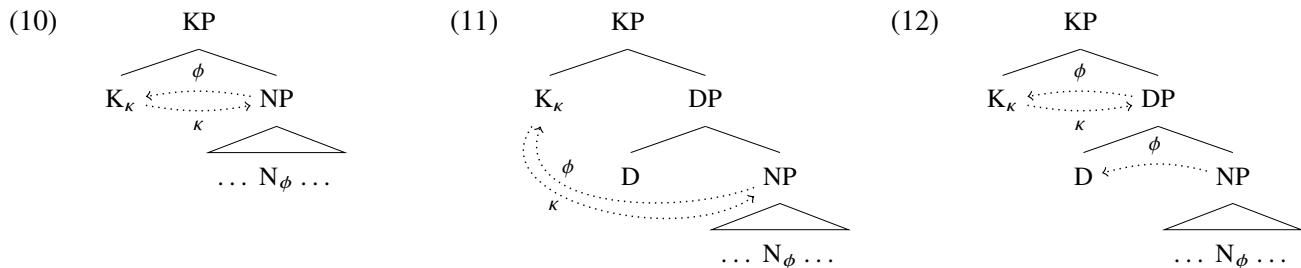
⁴In this paper, *nominal phrase* refers to any (extended) nominal projection, including DP and KP, not just an NP.

⁵Norris 2022: 6 notes that “KP has become so widely assumed that it has been used in published, peer-reviewed research without citation for over a decade. . .”.

3.2 Agree in Nominal Projections

What are the *minimal* assumptions that need to be made about Agree in maximal nominal projections headed by a case-bearing functional head, e.g., K? Given that 1) such maximal KP projections bear ϕ features (as is clear from standard binding and agreement facts), that 2) the source of ϕ features is N or projections just above it rather than K (a standard assumption), and that 3) there are no specialized feature percolation mechanisms (a methodological assumption we make here), K must Agree with its nominal complement in order to receive ϕ features. Then, since the K head is featurally identical to its projections,⁶ these ϕ features are also present on KP. Hence, K must be assumed to host a probe interacting with a c-commanded constituent bearing ϕ features; such a probe may be minimally specified as: [INT: ϕ , SAT: ϕ].

Assuming, as in the work on Agree summarized in §2 above, that probes flag their goals with their own (the probes') features, another result of this Agree is the transfer of K's case feature(s), κ , to its complement. This minimum exchange of features is schematically illustrated in (10), which assumes no DP layer.⁷



If a DP layer is present, there are two possibilities, as schematically shown in (11)–(12). The first is that D does not Agree with its NP complement, so the closest constituent bearing ϕ features that is c-commanded by K is the NP, with which K exchanges features directly (see (11)). In this case, the NP (and, hence, also its N head) ends up with the case features κ , but the DP (and, hence, the D) remains case-less. The second possibility is that D does Agree with NP, in which case the closest constituent bearing ϕ features that is c-commanded by K is the DP (see (12)). In this case, the DP (and, hence, the D), but not the NP (or the N), receives κ .

Note that none of these possibilities allows for concord; in all, exactly two categories bear κ : K(P) and either N(P) alone (in (10)–(11)) or D(P) alone (in (12)). Assuming that, if a feature of a head H is morphologically realized at all, it may be realized either on H itself or – as a phrasal affix – on H's maximal projection (but normally not both), up to two exponents of case in such structures are predicted, but also possibly one (for example, if K is phonologically empty and κ is not realized phrasally on KP) or even none (if category-specific case impoverishment rules are at play). An example of a language with nominal structures as in (12) may be English, where nominal agreement and case are realized only on (some) Ds (cf. *this great book* vs. *these great books* and *he* vs. *him*).

While there are many languages with just one or two possible exponents of case within any nominal projection, it is often not easy to decide which of the structures considered above are realized in a given language. This seems relatively clear in languages such as Diyari (Australian), where case is expounded on the determiner and on the final word of the nominal projection, whether it is a noun or an adjective, e.g., *nhu-lu karna pirna-li* 'he-ERG man big-ERG', i.e., 'the big man' in ergative (Austin 1981 apud Blake 2001: 101, (21)). Such languages probably have nominal structures as in (12), with κ expounded on D and – as a phrasal suffix – on KP. However, even here another possible analysis presents itself: the structure might be as in (11), with D head-moving to K and realizing the determiner–case bundle, KP realizing κ as a phrasal affix, and with nouns being morphologically impoverished for case. We leave such matters for language-specific research, but – for concreteness – we will assume that D normally Agrees with NP, as in (12), given that D expones the NP's gender and number in many familiar languages such as French or German.

We now move to the consideration of concord languages, in which case and ϕ features have a theoretically unbounded number of exponents within a single nominal projection.

⁶This follows from the common Minimalist assumption of Bare Phrase Structure (Chomsky 1995).

⁷Intrinsic features of particular heads are marked as subscripts.

3.3 Concord is Insatiable Agree

In concord languages, case and ϕ features are typically expounded on the noun and all agreeing modifiers, regardless of their number. This is illustrated with the following attested examples from Polish, each with five inflectionally adjectival modifiers⁸ agreeing with the noun in case (nominative in (13) and genitive in (14)) and in ϕ features.⁹

(13) [Jakiś drugi taki sam mały chłopiec] podskoczył. . .
 some.M.SG.NOM second.M.SG.NOM such.M.SG.NOM same.M.SG.NOM little.M.SG.NOM boy.M.SG.NOM jumped.M.3SG

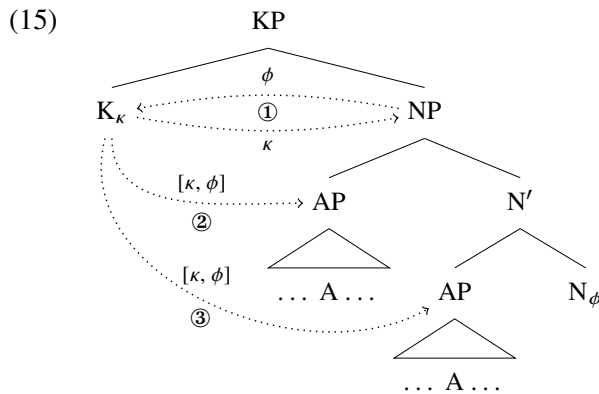
‘[Some little boy just like the other one] jumped. . .’ (POL)

(14) . . . spełnienie [tej swojej cichej wielkiej skrywanej namiętności]. . .
 fulfillment.N.SG.NOM this.F.SG.GEN self.F.SG.GEN quiet.F.SG.GEN great.F.SG.GEN hidden.F.SG.GEN passion.F.SG.GEN]

‘. . . fulfillment [of that quiet great hidden passion of her/his]. . .’ (POL)

In the previous subsection, the probe on K was assumed to be specified as $[INT:\phi, SAT:\phi]$. However, another possibility predicted by recent approaches to Agree is that the probe is insatiable: $[INT:\phi, SAT:-]$. We claim that it is this insatiable Agree that directly accounts for concord in languages such as Latin, Polish, and Estonian. To see this, recall Clem and Deal’s (2026) analysis of Shawi ergative (§2). There, a v probe first flags a constituent with its categorial v feature and “agrees” with that constituent in ϕ features, i.e., copies those features over. Subsequently, the same probe flags another constituent, but now not just with v , but with *all* its features – the whole $[v, \phi]$ bundle.

We propose that exactly the same mechanism is responsible for concord, the only difference being that K flags its nominal complement with its case, κ , rather than only with its category K. In the process, it copies over the ϕ features of the nominal complement, so that any subsequent probing involves flagging with the $[K, \kappa, \phi]$ bundle.¹⁰ This way not only the particular case κ , but also the noun’s ϕ features end up on all accessible constituents within the maximal nominal projection. This is schematically illustrated in (15), with two adjectival modifiers.¹¹ Once this tree is fully built, all constituents contain both ϕ and κ among their features.



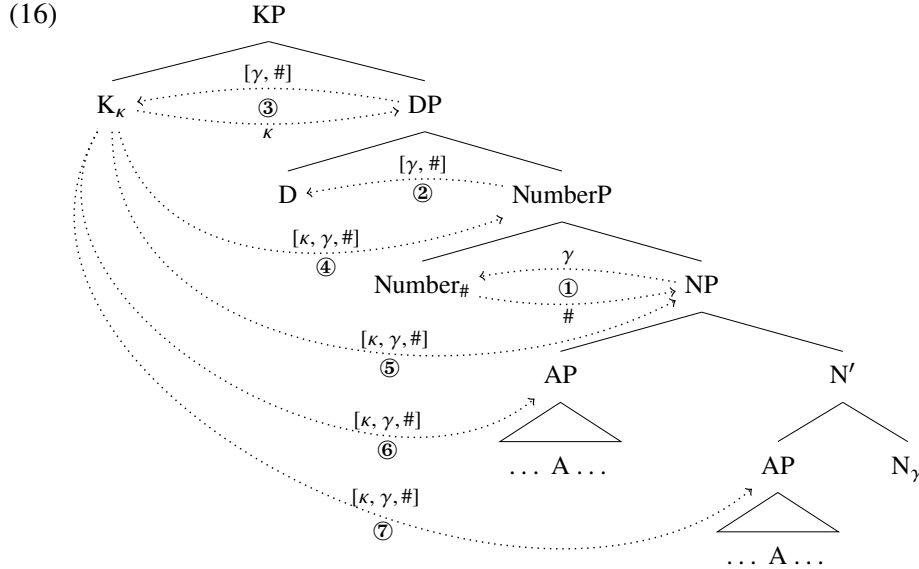
⁸By “inflectionally adjectival modifiers” we understand not only standard APs, but also, for example, (phrases headed by) adjectival participles, as well as demonstrative pronouns, which may or may not be reasonably analyzed as D(eterminer)s, but which, like prototypical adjectives, inflect for case, gender and number in languages such as Polish.

⁹Unless explicitly indicated otherwise, Polish examples are from the (pre-LLM) National Corpus of Polish (NCP; Przepiórkowski *et al.* 2012; <https://nkjp.pl/>); some are slightly simplified in ways that do not affect corresponding claims.

¹⁰A somewhat similar mechanism is assumed – but not fully formalized – in Pesetsky 2013: 32. Since category flagging does not matter for most of this paper, we will usually omit categories in such feature bundles. See Philippova 2018 and Przepiórkowski 2026 for a phenomenon that may be analyzed as category flagging expounded independently of case flagging.

¹¹We make here a simplifying assumption that Agree only targets maximal projections, e.g., NP but not N' or N in (15). (Technically, this can be ensured by requiring that any given probe does not target the same goal twice and that constituents with the same – shared – labels count as the same goals.) However, nothing seems to hinge on this assumption. In its absence, Agree would also target non-maximal projections, attempting to flag them with a feature bundle that they already have (via identity with the higher projection that has already been flagged). If feature bundles do not constitute sets but rather stacks, as assumed below, this might lead to the presence of multiple identical feature bundles on a label. However, such bundles would never be multiply expounded, given the widely assumed haplology of affixes of the same type (see, e.g., Richards 2013: 50–51), so their presence on the stack does not seem to be a problem. Nevertheless, if this multiple presence of identical bundles turns out to be technically problematic in some way, a syntactic “feature bundle haplology” rule may be assumed, which prohibits adding to the stack a bundle that is already present on top of the stack.

In (15), the only functional head is K, but concord works similarly in more articulated structures. This is illustrated in (16), which includes a DP layer and assumes that grammatical number, #, is introduced by a separate functional head (Ritter 1991), while the lexical N is intrinsically specified for grammatical gender, γ . (This structure could be further extended by splitting lexical N into root and nominalizer, by including a separate functional head introducing grammatical gender, etc.)



Here, in the first interesting step, Number Agrees with the NP, resulting in bidirectional passing of relevant ϕ features: γ from NP to Number and # from Number to NP. At this point Number contains both gender and number, and – hence – so does its maximal projection. Then, D Agrees with NumberP, copying over its ϕ features γ and #. In the third relevant step, K Agrees with the DP, resulting in the strengthening of its κ flag to $[\kappa, \gamma, \#]$. In further steps, K insatiably Agrees with other constituents it c-commands. As a result, all constituents bear the $[\kappa, \gamma, \#]$ bundle, resulting in concord.¹²

The question may arise at this point about the featural composition of AP goals that attract $[\text{INT}:\phi, \text{SAT}:-]$ probes. We adopt the standard assumption that such probes interact with nodes that bear valued ϕ features, but we extend it to also include nodes that bear unvalued ϕ features.¹³ In languages such as Polish, where adjectives inflect for case, number, and gender, they are lexically specified as *u*CASE, *u*NUMBER, and *u*GENDER, which indicates that these features may be assigned specific values in syntax. By comparison, nouns may be assumed to bear specific gender values already in the lexicon, e.g., feminine, and perhaps a specific number (especially, *plurale tantum* nouns), but they are also lexically specified for unvalued *u*CASE, while adverbs do not bear any CASE, GENDER or NUMBER specification in the lexicon and, hence, may never receive values of these features in syntax. Given that $\text{INT}:\phi$ probes interact with any nodes that bear ϕ features, they will interact both with nouns (some of whose ϕ features are valued) and with adjectives (whose ϕ features are not valued lexically), but not with adverbs, which do not bear ϕ features at all (neither valued nor unvalued). In the case of valued ϕ features on the goal, these values are copied to the probe. However, unvalued ϕ features have no values to be copied, so the effect of Agree with nodes containing only unvalued features is pure flagging by the probe, without any feature copying from the goal to the probe, as illustrated in the two trees above.

The assumption that adjectives are lexically specified as *u*GENDER, *u*NUMBER and *u*CASE, combined with standard feature geometries (Harley and Ritter 2002, Caha 2009) and with the now common assumption that unvalued features do *not* cause derivation failure (e.g., Preminger 2014), makes the following prediction. Given that GENDER, NUMBER

¹²Technically, Number(P) and N(P) also bear the earlier bundle $[\gamma, \#]$, unless some kind of “feature bundle haplology” rule (see fn. 11) is assumed. This is not a problem, given the common mechanisms of expounding the outermost (most recent) or the most marked feature bundle, discussed in §4.

¹³Recall that (un)valuedness is dissociated from (un)interpretability, as the latter is not a factor in modern Agree (Deal 2025).

(17) Jasne było, [że pochodzi on z kradzieży].
clear.N.SG.NOM was.N.3SG that comes.3SG M.3SG.NOM from theft
'It was clear that it had been stolen.' (POL)

3.4 Non-Agreeing Constituents

(18) Nie poddawajcie się [tej zgubnej porze roku].
 NEG surrender.IMP.2PL this.F.SG.DAT fatal.F.SG.DAT season.F.SG.DAT year.M.SG.GEN
 ‘Do not surrender to [this fatal time of year.M.GEN].F.DAT.’ (POL)

(19)

Diagram illustrating a syntactic tree structure for a probe hierarchy, showing the relationship between various nodes and their associated features.

The tree structure is as follows:

- KP** (Root) branches into **K_K** and **DP**.
 - K_K** is linked to **DP** via a dotted arrow labeled ③ and ϕ .
- DP** branches into **D** and **NP**.
 - D** is linked to **NP** via a dotted arrow labeled ② and ϕ .
- NP** branches into **AP** and **N'**.
 - AP** is linked to **N'** via a dotted arrow labeled ④ and $[\kappa, \phi]$.
- N'** branches into **N _{ϕ}** and **KP** (dashed circle, labeled "horizon for the probe on the higher K").
 - N _{ϕ}** is linked to the lower **KP** via a dotted arrow labeled ⑤ and $[\kappa, \phi]$.
- The lower **KP** branches into **K_{GEN}** and **DP**.
 - K_{GEN}** is linked to the lower **DP** via a dotted arrow labeled ① and M.SG / GEN .
- The lower **DP** branches into **N_{M.SG}**.

Additional features and probes are indicated by dotted arrows and labels:

- ⑥** and **⑦** are marked with a red **X**, indicating failed probes.
- ⑥** is linked to **AP** via a dotted arrow and $[\kappa, \phi]$.
- ⑦** is linked to **N _{ϕ}** via a dotted arrow and $[\kappa, \phi]$.

¹⁴While constituents properly dominated by the horizon XP are clearly beyond the reach of the probe on the c-commanding X, the phenomena considered in Keine 2019, 2020 do not determine whether XP itself is reachable by the probe. Concord provides an argument that it is not. If it were, the higher K would endow the embedded KP with its ϕ features and copy over the ϕ features of that KP, with disastrous effects for agreement and binding.

this assumption, as long as this KP is anywhere inside the higher KP. In any case, this lower KP will turn out to be a horizon for the probe on the higher K. The ensuing steps two through five are exactly as explained in §3.3 above. What is new here is that, in the attempted steps six and seven, interaction of the probe on the higher K with the lower KP or anything within it is impossible, as KP is that probe's *horizon*.¹⁵

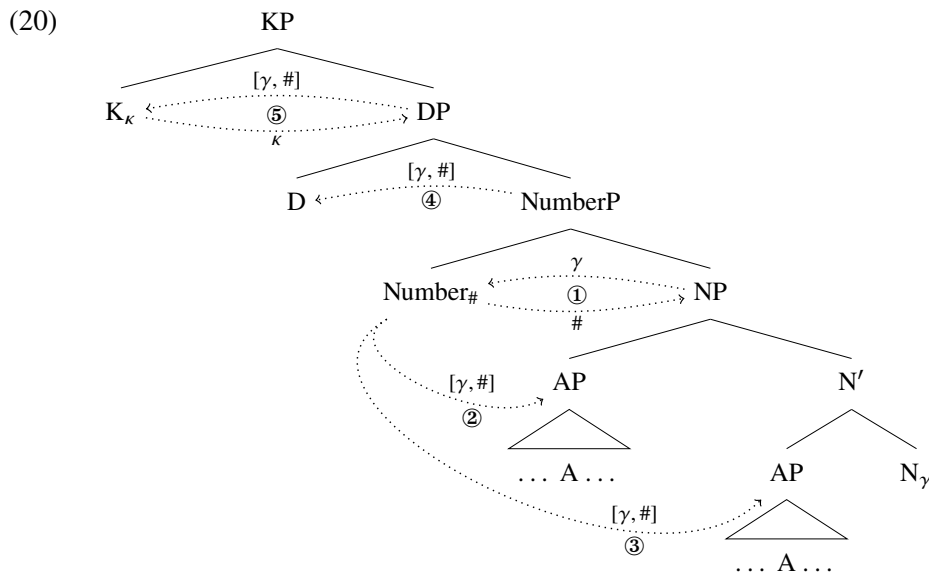
Note that, while both satisfaction specifications and *horizons* constrain the search space (Deal 2025: 23, 27, Deal 2026: 242), they are conceptually very different mechanisms with rather different outcomes. When a constituent is encountered that matches a SAT specification, interaction with this constituent is possible, but then search for goals stops. By contrast, when a *horizon* is encountered, interaction with it or anything within it is not possible, but the probe may continue to search for other goals elsewhere.

3.5 Typology of Concord

Given that the KP projection is commonly taken to be the maximal nominal projection, certainly above the Number projection, the proposed approach to concord makes a typological prediction: in languages with both overt case and overt ϕ features, concord may target either both kinds of features, or ϕ features alone, but not case alone. This is why. In §3.3, we saw how an insatiable probe on K results in full concord in case and ϕ features. If there is no insatiable probe, there is no concord, as discussed in §3.2. However, there is also another possibility, namely, that the probe on K is immediately satiable, but the one on D – or Number, if present – is not. Then, only ϕ concord ensues.

Let us illustrate this possibility with (20), i.e., a structure just like (16), but with an insatiable probe present on Number instead of K. (Essentially the same reasoning applies if the insatiable probe is present on D, and if only one of the DP and NumberP layers is present.) As in (16), in the first interesting step, Number Agrees with the NP, resulting in bidirectional passing of relevant ϕ features: γ from NP to Number and $\#$ from Number to NP. However, since the Number probe is insatiable here, it flags both APs with γ and $\#$ in the subsequent two steps, resulting in ϕ concord. The final two steps are identical to the corresponding two steps in (16), but – unlike there – they are not followed here with K flagging all maximal projections with $[\kappa, \gamma, \#]$, as the K probe is now immediately satisfied when it Agrees with the DP. This way, κ is only present on D(P) – and may be exponed there – but not on APs or directly on the NP.

What the current system cannot derive is the opposite situation, i.e., case concord with no ϕ concord. This is because, if there is case concord, the probe on K must be insatiable, but if it is, it should flag all constituents not only with case, but also with the ϕ features it copied from its nominal sister in the first probing step.



¹⁵Given that *horizons* are an addition to – not a replacement of – *phases* (Keine 2019, 2020), step 7 is also prohibited by the standard (strong) Phase Impenetrability Principle (PIC) of Chomsky 2000, 2007, 2008, on the common assumption that maximal nominal projections are phases. See, e.g., Citko 2014: §4.3 and references therein on DP *phases* (on the assumption that DPs are maximal nominal projections), as well as, e.g., Richards 2010, Norris 2014, and Levin 2015, for explicit analyses of KPs/PPs as *phases*.

Typological considerations confirm the correctness of this prediction. As discussed in Bayırlı 2017: ch. 2, there are many languages with complete case and ϕ concord and multiple languages with pure ϕ concord despite the presence of a case system, but no known languages with pure case concord despite the presence of overt ϕ exponents.¹⁶ Importantly, no additional assumptions are needed to derive this typological generalization, apart from the independent and universally adopted assumption that KP is the maximal nominal projection, higher than any projections introducing ϕ features.^{17,18}

3.6 Interim Summary

The theoretical advantage of the analysis of concord proposed in this section over previous approaches (e.g., Norris 2014 and Bayırlı 2017) is that it does not posit any special feature percolation principles and does not rely on any non-standard locality assumptions. Instead, it relies on relatively recent but already widely adopted approaches to Agree, on which a single probe may target multiple goals, flagging each with its features, as well as on an also relatively recent but amply independently motivated *horizons*-based approach to locality. Empirically, this analysis of concord predicts out of the box not only the most standard concord facts, but also certain typological generalizations.

In the following sections, this approach is stress tested by applying it to heterogeneous case and ϕ patterns in Slavic and Finnic.

4 Heterogeneous Patterns in Slavic

4.1 Introduction

A well-known feature of north (i.e., East and West) Slavic numeral–noun constructions (NNCs) is the contrast between their heterogeneous case patterns in direct cases (NOM, ACC) and homogeneous patterns in oblique cases (GEN, DAT, LOC, INS) – see, e.g., Mel’čuk 1985 and Babby 1987 on Russian, and Franks 1994, 1995 on Slavic more generally. This is illustrated on the basis of Polish data in (21).

(21)	position	‘good.A’	‘books.N’	‘five.NUM’	‘books.N’	‘mass.N’	‘books.N’
	NOM	<i>dobrze.NOM</i>	<i>książki.NOM</i>	<i>pięć.NOM</i>	<i>książek.GEN</i>	<i>masa.NOM</i>	<i>książek.GEN</i>
	ACC	<i>dobrze.ACC</i>	<i>książki.ACC</i>	<i>pięć.ACC</i>	<i>książek.GEN</i>	<i>masę.ACC</i>	<i>książek.GEN</i>
	GEN	<i>dobrych.GEN</i>	<i>książek.GEN</i>	<i>pięciu.GEN</i>	<i>książek.GEN</i>	<i>masy.GEN</i>	<i>książek.GEN</i>
	DAT	<i>dobrym.DAT</i>	<i>książkom.DAT</i>	<i>pięciu.DAT</i>	<i>książkom.DAT</i>	<i>masie.DAT</i>	<i>książek.GEN</i>
	LOC	<i>dobrych.LOC</i>	<i>książkach.LOC</i>	<i>pięciu.LOC</i>	<i>książkach.LOC</i>	<i>masie.LOC</i>	<i>książek.GEN</i>
	INS	<i>dobrymi.INS</i>	<i>książkami.INS</i>	<i>pięcioma.INS</i>	<i>książkami.INS</i>	<i>masq.INS</i>	<i>książek.GEN</i>

The first two columns contain Polish for ‘good books’ in various case positions.¹⁹ As expected of a full concord language, the adjective agrees with the noun in case, number, and gender. The final two columns contain Polish for ‘a mass of books’, headed by the singular noun *masa* ‘mass’. Again, as expected, the nominal complement of this noun is in the genitive in all case positions. By contrast, the Polish numeral–noun construction for ‘five books’ in the middle columns (in bold) behaves like the noun–noun construction in NOM and ACC, where the noun is in the genitive, and like the adjective–noun construction in the other positions, where the noun agrees with the numeral

¹⁶Note that it is possible that a language has grammatical case but no grammatical number or grammatical gender (just as, e.g., Turkish has no grammatical gender), in which case pure case concord is a possibility predicted by our proposal.

¹⁷By contrast, these – and more – typological generalizations are derived in Bayırlı 2017 with the help of specific feature percolation principles and non-standard assumptions about phases.

¹⁸Similarly, given that gender-introducing heads are closer to N than number-introducing heads (see, e.g., Saldana *et al.* 2025 and references therein), the possibility of concord in gender – to the exclusion of number and case – is predicted. This prediction seems to be correct at least for Shughni (a Pamir language spoken in Tajikistan, Afghanistan, Pakistan, and China) and Bangla (an Indo-Aryan language of Southeast Asia); see Bayırlı 2017: 31–33.

¹⁹Apart from these 6 cases, Polish also has the vocative case used for address and not selected by any predicates. As such, it may be regarded as being outside the case system (cf. Blake 2001: §1.2.3) and will be ignored in the rest of this paper, as is common in discussions of Slavic numerals. (Vocative NNCs always have the same form as nominative in Polish, as vocative always equals nominative in the plural and NNCs are always plural.)

in case.^{20,21} Similar patterns of NNCs are observed in other West Slavic languages (including Czech) and in East Slavic (including Russian and Ukrainian), but not in South Slavic.

4.2 Analysis

4.2.1 Idea

The main idea of the following account is based on that of Pesetsky 2013 (which in turn was based on Richards 2013), but the analysis proposed below is free of the considerable number of controversial assumptions made there and it attempts to be more explicit and precise.

The basic idea of Richards 2013 and Pesetsky 2013 is that Russian is a case stacking language, but only the last assigned – i.e., the outermost – case is overtly realized.²²

(22) *Expone Outermost*: Expone only the outermost feature bundle. (cf. Pesetsky 2013: 11, (7))

In oblique positions, externally assigned case percolates down and ends up on nominals previously marked as GEN, resulting in case stacking. But given the *Expone Outermost* rule in (22), only the oblique case is expounded. This approach is equally applicable to Polish and it accounts for the GEN, DAT, LOC, and INS rows in the table in (21). However, if nothing more is said, the GEN case on the noun in the NOM and ACC rows is unexpected: it should be NOM and ACC, respectively. In order to prevent the spread of NOM and ACC to the genitive noun, a relatively large number of non-standard assumptions are made in Pesetsky 2013.²³

We modify this general covert case stacking idea by following the insight in Grishin 2024 that there are phenomena which call for a different exponence rule than (22), namely, a rule like (23):²⁴

(23) *Expone Most Marked*: Expone only the most marked feature bundle.

Such a rule seems to be at play in Gothic free relatives, as shown in (24) (Harbert 1992: 111 apud Grishin 2024: 47):²⁵

- (24) a. [Pan=ei frijos] siuks ist.
 who.ACC=C you.love_{ACC} sick is_{NOM}
 ‘(The one) whom you love is sick.’ (GOT)
- b. [Po=ei ist us Laudeikaion] jus ussiggwaid.
 which.ACC=C is_{NOM} out Laodicea you read_{ACC}
 ‘And read (the one) which is from Laodicea.’ (GOT)

Unlike in contemporary German and many other languages, where the relative pronoun must simultaneously satisfy case requirements of both the matrix and the embedded clause, in Gothic the relative pronoun occurs in the more marked case of the two required in the two clauses. For example, in (24a), where ACC is expected in the relative clause and NOM in the matrix clause, the pronoun occurs in ACC. By contrast, in (24b) it is the matrix clause that expects ACC, with the relative clause expecting NOM, but the pronoun is still in ACC.

This is expected if the *Expone Most Marked* rule in (23) is at play, on the assumption that ACC is more marked than NOM. But this assumption is independently and abundantly justified, as argued at length – on the basis of

²⁰In the genitive position, it is not immediately clear whether GEN on the noun is assigned as in nominative and accusative positions, or whether it agrees with the numeral as in other oblique positions. Russian paucal numerals (§5) and pseudopartitive constructions in Finnic (§6) show that GEN patterns with other oblique cases there, so we assume it also does in Polish, as described in the main text.

²¹Note that the numeral in the NOM row is marked in gray, as there are good arguments for analyzing Polish numeral subjects as actually in the accusative (see, e.g., Schenker 1971, Franks 1994, 1995, Przepiórkowski 1996, 1999, 2004, Miechowicz-Mathiasen 2012, Willim 2015, Witkoś *et al.* 2018, Citko *et al.* 2018). The question of the source of this accusative is outside the scope of this paper.

²²The rule in (22) is renamed from the original *One-Suffix Rule* for added precision and generalized for compatibility with our set of assumptions. While in Pesetsky 2013 it is only applied to stacked cases in Russian, Grishin 2024 extends it to stacked ϕ -feature bundles in Passamaquoddy-Wolastoqey (Eastern Algonquian; Canada and USA).

²³These include some very special properties of null Determiners resulting in their selective assignment of the nominative case, an analysis of those accusative objects which are (commonly taken to be) syncretic with nominative as *actually* NOM, some very special structures of prepositional phrases (apparently) headed by prepositions assigning ACC or GEN, etc. (cf. Przepiórkowski 2026).

²⁴A similar idea applied to Russian cardinals, which foreshadows the analysis proposed below, may also be found in Matushansky 2008: §6 (a proceedings paper we became aware of after developing the analysis presented here).

²⁵A subscripted case value indicates case expectations of a given word (or an associated functional head).

a number of phenomena, including case syncretisms – in Caha 2009, where the following hierarchy of morphological cases is proposed (with the least marked cases on the left):

(25) NOM < ACC < GEN < DAT < INS < . . .

Given this independently motivated case hierarchy, the *Expone Most Marked* rule immediately explains the puzzling case patterns in Slavic NNCs, as shown in (26).²⁶

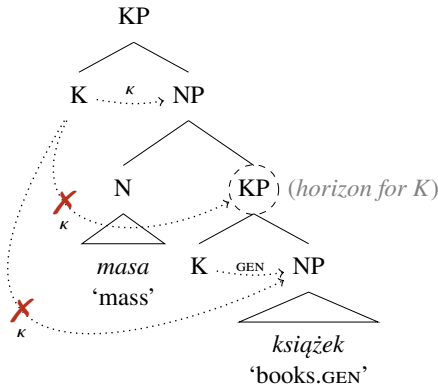
(26) position	‘five.NUM’	‘books.N’
NOM	<i>pięć</i> .NOM	<i>książek</i> . GEN .NOM
ACC	<i>pięć</i> .ACC	<i>książek</i> . GEN .ACC
GEN	<i>pięciu</i> .GEN	<i>książek</i> . GEN . GEN
DAT	<i>pięciu</i> .DAT	<i>książkom</i> . GEN . DAT
LOC	<i>pięciu</i> .LOC	<i>książkach</i> . GEN . LOC
INS	<i>pięcioma</i> .INS	<i>książkami</i> . GEN . INS

This table repeats the middle columns of the table in (21), but with case stacking shown explicitly, and with the most marked case in each stack boxed. It is exactly the boxed cases that are expounded, as predicted by the *Expone Most Marked* rule in (23).²⁷

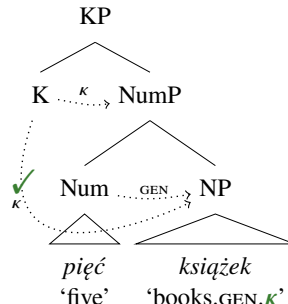
4.2.2 Structures of NNCs

Despite the vast literature on Slavic numeral–noun constructions, there is still no agreement on almost any aspect of their syntax. However, the one aspect that is crucial for the analysis below is currently generally accepted: numerals occur within an extended nominal projection, rather than introducing a separate nominal projection. In our terms, this means that, while noun–noun constructions involve two KPs, ultimately headed by the two nouns, numeral–noun constructions involve a single KP ultimately headed by the single noun. This is schematically shown in (27)–(29) (for perspicuity, without DP or any other layers).

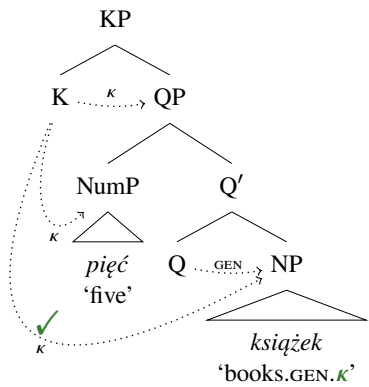
(27) noun–noun construction:



(28) NNC (Num is head):



(29) NNC (NumP is in Spec,QP):



In (27), corresponding to *masa książek* ‘(a) mass (of) books’, the lower KP (and the NP within it) is immune to the probe on the higher K, as it is (within) a horizon for that probe, as discussed in §3.4 above. So, whatever case the higher K bears, it only affects the higher noun, *masa* ‘mass’, and not the lower noun, *książek* ‘books.GEN’.

The trees in (28)–(29), by contrast, correspond to the two most common approaches to Slavic numeral–noun constructions. On the first, illustrated in (28), numerals are heads in extended nominal projections, and they directly assign the genitive case to their nominal complements. This kind of structure is assumed for Polish in Klockmann 2017 and – for various Slavic languages – in countless earlier works.²⁸ On the second approach, illustrated in (29),

²⁶Recall (from fn. 21) that, in Polish, numeral subjects are actually ACC, not NOM – hence the gray NOM in (26). However, in Russian and other Slavic languages, such numeral subjects may occur in the nominative, and the argument in the text is valid whether the case assigned to numeral subjects is NOM or ACC, as both are less marked than GEN.

²⁷In the GEN position, there are two GEN feature bundles on the noun. In §5 we will see that it is the outermost of these that is expounded.

²⁸However, in Klockmann 2017, it is not Num directly but rather dependent case mechanisms (not made precise there) that are responsible for the genitive on the noun.

numerals are specifiers of empty quantificational heads in extended nominal projections, and it is such empty heads, Q, that are responsible for the genitive on the noun. Such structures are assumed for Polish in Witkoś *et al.* 2018 and – for various Slavic languages – in multiple earlier works.

As shown in (28)–(29), whichever of the two popular approaches to the structure of NNCs is adopted, case stacking is predicted, and for the same reason: there is no horizon (or phase) boundary between the K and the genitive NP that would block flagging the NP with κ . Hence, the noun ends up with two cases, GEN and κ , and expones GEN when κ is the less marked NOM or ACC; otherwise, it expones the κ , as the most marked case on the stack. This analysis remains valid when more layers in extended nominal projections are assumed, e.g., also DP and/or NumberP, as long as these additional layers do not constitute additional locality boundaries.

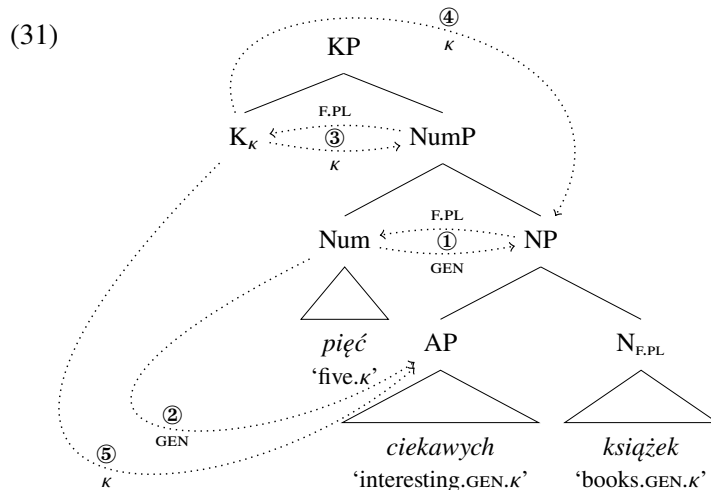
While the analysis presented below is compatible with both NNC structures (28)–(29), it is illustrated only with the simpler NNC structure in (28) for reasons of space.

4.2.3 Insatiable Probes on Num

We argued in §3.3 that, in concord languages, case-flagging probes on K are insatiable. In this section, we show that, in north Slavic NNCs, probes on Num are also normally insatiable. (We will revise this statement in §5.2.2, where we show that a successful analysis of Russian paucal numerals also requires an immediately satiable probe.)

The reason for postulating insatiable probes on Num becomes clear when NP contains an AP, as in (30).

- (30) pięć {ciekawych / *ciekawe} książek
 five.F.NOM/ACC interesting.PL.GEN / *NOM/ACC book.F.PL.GEN
 ‘5 interesting books’ (POL)



When such a numeral phrase appears in nominative and accusative positions, the adjective must occur in the genitive, like the noun, and not in the nominative/accusative, like the numeral. Now, if the probe on Num were not insatiable, it would only flag the NP with GEN, as shown in step one in (31). If so, the AP within that NP would at this stage be case-less, and it would only receive the κ case from the higher K, here, the NOM/ACC. This would predict the grammaticality of the variant of (30) with the form *ciekawe* ‘interesting.NOM/ACC’, contrary to facts. However, if the probe on Num is insatiable, it flags the AP with GEN in step two. Then, in steps three to five, all constituents within the KP are flagged with κ by K, resulting in the stacked GEN. κ not only on the noun, but also on the adjective, correctly exponed as GEN when κ is less marked (and as κ when it is more marked than GEN).

In summary, the probes on K and Num are very similar: they are both insatiable and both involve flagging with a case value, as made explicit in (32)–(33).

- (32) probe on K_κ : [INT: ϕ , SAT:–, FLAG: κ]
 (33) probe on Num: [INT: ϕ , SAT:–, FLAG:GEN]

4.3 Something Interesting

According to the above analysis, the crucial property of Slavic NNCs that results in the heterogeneous case pattern illustrated in the table in (26) is the presence of another – apart from K – case-assigner within the extended nominal projection, namely, the Numeral (or Q, if the NNC structure in (29) is assumed). This predicts that, in principle, there might be other – non-numeral – constructions in which an item on the nominal spine idiosyncratically assigns GEN, resulting in the same heterogeneous case pattern. This prediction is borne out.

There are a handful of Polish pronouns, all containing the root *c(o)*, which display a heterogeneous vs. homogeneous case pattern similar to that of numerals; these are: *co* ‘what’, *coś* ‘something’, *cokolwiek* (and its colloquial/archaic variant *cośkolwiek*) ‘whatever’, and *nic* ‘nothing’. This case pattern is illustrated in (34) (the bolded initial columns) on the basis of *coś ciekawego* ‘something interesting’.

(34)	position	‘something interesting’		‘somebody interesting’	
	NOM	<i>coś</i>.NOM	<i>ciekawego</i>.GEN	<i>ktoś</i> .NOM	<i>ciekawym</i> .NOM
	ACC	<i>coś</i>.ACC	<i>ciekawego</i>.GEN	<i>kogoś</i> .ACC	<i>ciekawego</i> .ACC
	GEN	<i>czegoś</i> .GEN	<i>ciekawego</i> .GEN	<i>kogoś</i> .GEN	<i>ciekawego</i> .GEN
	DAT	<i>czemuś</i> .DAT	<i>ciekawemu</i> .DAT	<i>komuś</i> .DAT	<i>ciekawemu</i> .DAT
	LOC	<i>czymś</i> .LOC	<i>ciekawym</i> .LOC	<i>kimś</i> .LOC	<i>ciekawym</i> .LOC
	INS	<i>czymś</i> .INS	<i>ciekawym</i> .INS	<i>kimś</i> .INS	<i>ciekawym</i> .INS

These few pronouns are the sole exception to the rule that adjectival phrases within nominal phrases always agree in case with the nominal heads in Polish; as shown in (34), in NOM and ACC positions APs modifying *coś* ‘something’ occur in GEN, instead of the expected NOM or ACC. In other case positions, APs occur in the expected agreeing case, mirroring the distribution of NPs in NNCs. This behavior should be contrasted with the usual situation, exemplified in the final columns of (34), which illustrate that pronouns based on the root *kt(o)*, e.g., *ktoś* ‘somebody’, do not display this case pattern; here APs always agree with the pronoun, as they do elsewhere in Polish.

We do not attempt to explain why exactly *coś* ‘something’, but not *ktoś* ‘somebody’, flags its adjectival modifiers with GEN.²⁹ But, given that it does, the pattern shown in (34) is fully expected on the approach to concord presented above: since AP is not (within) a horizon for the higher K (and not within a phase), this K may flag this AP with case κ , on top of GEN that this AP receives from *coś*, resulting in case stacking and the realization of the most marked case. On this account, the only idiosyncrasy of the Polish pronominal root *c(o)* is that it flags the AP with GEN; the mixed pattern seen in (34) follows from the lack of any locality boundary between the K and the AP and from the *Exponere Most Marked* rule in (23).

This is confirmed by the fact that a genitive KP dependent of *coś* ‘something’ must stay in the genitive, regardless of the external case. While such an additional genitive nominal dependent of *c(o)* is often awkward due to the structural ambiguity it may create,³⁰ it is relatively easy to find occurrences in corpora that exemplify this pattern, e.g., (35) from the National Corpus of Polish.

- (35) Chciałbym dla odmiany przeczytać coś lżejszego Twojego autorstwa. . .
 want.M.1SG.SBJV for change read.INF something.ACC lighter.GEN your.GEN authorship.GEN
 ‘I’d like to read something lighter by you, for a change.’ (POL)
 (lit.: ‘... something.ACC lighter.GEN [your authorship].GEN’)

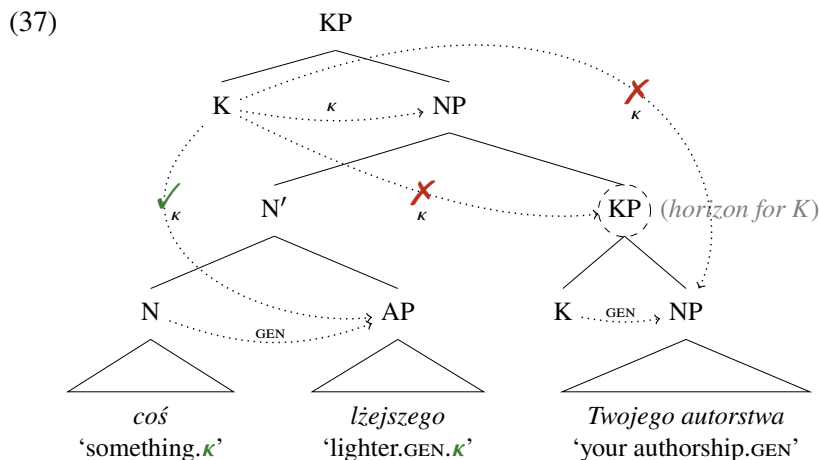
In (35) the *coś* phrase is in an ACC position, so not only the KP *Twojego autorstwa* ‘your authorship’, but also the AP *lżejszego* ‘lighter’ is in GEN. By contrast, when such a phrase occurs in an oblique (e.g., LOC) position, *coś* ‘something’ and the AP occur in the oblique case, as discussed above, but the KP remains in GEN, as shown in the constructed (36).

²⁹In terms of Klockmann 2017, the root *c(o)* – but perhaps not *kt(o)*, etc. – would be analyzed as idiosyncratically semi-lexical, in the sense that it hosts certain functional features (here, a GEN-flagging probe); see also Rutkowski and Szczegot 2001 for a similar analysis.

³⁰For example, *coś ciekawego Szekspira* ‘something interesting.GEN Shakespeare.GEN’ may be understood as ‘something interesting/curious by Shakespeare’, or as ‘something by the curious/interesting Shakespeare’.

- (36) Porozmawiajmy o czymś lżejszym Twojego autorstwa.
 talk.1PL.IMP about something.LOC lighter.LOC your.GEN authorship.GEN
 ‘Let’s talk about something lighter by you.’ (POL)
 (lit.: ‘... something.LOC lighter.LOC [your authorship].GEN’)

The schematic structure of such nominal phrases is given in (37), without commitment to any particular position of the AP and the inner KP; what is crucial is only that the inner KP is a horizon, so it – and the NP within it – is not accessible to the probe on the higher K, while there is no similar locality boundary between the higher K and the AP.



In summary, the “something interesting” idiosyncrasy – usually ignored in analyses of Polish mixed case patterns – confirms the general approach to concord and the particular analysis of Slavic numeral–noun constructions offered above. Apart from the fact that Numerals (or Qs) are case-assigners, there is nothing special about numeral constructions that gives rise to the famous heterogeneous vs. homogeneous case pattern in NNCs: this pattern follows from the *Expone Most Marked* rule and an independently motivated theory of locality, just as it does in the “something interesting” construction.³¹

4.4 Interim Summary

On the proposed approach to concord, the famous Slavic heterogeneous vs. homogeneous pattern in Slavic numeral–noun constructions falls out from general and independently needed principles. It follows directly from insatiable Agree, which results in feature bundle stacking, and from the *Exponere Most Marked* rule, needed independently to account for Gothic free relatives, among other phenomena. The only idiosyncrasy is that Slavic numerals – or the Q heads associated with them – flag their dependents with GEN, an idiosyncrasy long accepted in Slavic linguistics and shared with some indefinite pronouns.

In the next section, we show how this approach to concord and north Slavic NNCs naturally extends to Polish and Russian paucal numerals, one of the most baffling syntactic phenomena in Slavic syntax.

5 Paucal Numerals in Slavic

In a number of Slavic languages, the syntax of paucal numerals – typically expressing ‘two’, ‘three’ and ‘four’ – is more complex than that of higher numerals. In Proto-Slavic, words expressing numbers 2–4 were agreeing adjectival modifiers, while 5–10 were feminine nouns that combined with genitive plural NPs. The emergence of the currently widely recognized separate “numeral” grammatical category was triggered by the decay of dual number in north Slavic during the Middle Ages. However, as demonstrated in Nessel 2020, the system of Russian paucal numerals has

³¹It might seem more attractive to propose an account on which the relevant pronouns involve some covert numeral that is responsible for the mixed case pattern in the “something interesting” construction. However, as noted in Rutkowski and Szczegot 2001, some Lithuanian pronouns, including *kažkas* ‘something’, behave like Polish *co* ‘something’, even though Lithuanian NNCs do not display the Slavic heterogeneous vs. homogeneous case pattern, which makes a numeral-based analysis of “something interesting” unmotivated.

continued to undergo radical change over the last 200 years. Given this unstable nature of Slavic paucals, a certain amount of idiosyncrasy in lexical entries of paucal numerals is to be expected.

The aim of this section is to provide an account of Polish and Russian paucal numerals that conservatively extends the approach to concord sketched above: while it was sufficient to only consider case stacking above, we will have to consider the stacking of larger feature bundles – similar to those considered in Clem and Deal 2026 (recall (3)–(5) in §2 above) – to account for Russian paucals. Moreover, an extension of the *Exponere Most Marked* rule will be proposed that will still handle the constructions considered above, but will also account both for Russian paucals considered in §5.2 and for Finnic pseudopartitives discussed in §6. We start with the simpler case of paucal numerals in Polish.

5.1 Paucals in Polish

One aspect of the complexity of paucal numerals in Polish and Russian is that the syntactic structure of the paucal construction depends to some extent on the grammatical gender of the noun. In Polish, it makes sense to distinguish 5 genders (Mańczak 1956): feminine (F), neuter (N), masculine-inanimate (M3), masculine-animate (M2), and masculine-human (M1). The names of the three masculine genders suggest a semantic animacy distinction, but the correlation of M2 with animacy is very imperfect: there are many inanimate M2 nouns (including all masculine names of car makes and dances, e.g., *Fiat* and *walc* ‘waltz’), and even some M2 nouns with a feminine natural gender denotation (e.g., *babsztyl* ‘woman.DEROGATORY’).

The syntax of higher numeral constructions does not depend on gender; as discussed in the previous section, the noun must occur in GEN when the NNC is in a NOM or ACC position, regardless of the noun’s gender. The situation is different in the case of paucal numerals, as shown in (38)–(39). Here, only in M1 is the typical numeral syntax observed, with the noun in the genitive (see (38)). By contrast, “adjectival” syntax is possible in all genders (including M1), where both the numeral and the noun occur in the case specific to a given syntactic position, e.g., in NOM (see (39)).³²

- (38) dwóch ochroniarzy
two.M1 guard.M1.PL.GEN
‘2 (male) guards’ (POL)
- (39) a. dwie lampy
two.F.NOM lamp.F.PL.NOM
‘2 lamps’ (POL)
- b. dwa {okna / stoły / psy}
two.N/M3/M2.NOM window.N.PL.NOM table.M3.PL.NOM dog.M2.PL.NOM
‘2 {windows / tables / dogs}’ (POL)
- c. dwaj ochroniarze
two.M1.NOM guards.M1.PL.NOM
‘2 (male) guards’ (POL)

In terms of the current account, the main difference between higher numerals and paucal numerals is that the former uniformly assign the genitive, which gets exponed as the more marked case in NOM and ACC positions, while most paucals do not assign GEN, so there is no competition and the nominative case is exponed in (39).

A natural analysis of this contrast is in terms of slightly different probes on the relevant numerals, shown below with explicit information about flagging and selectional restrictions:³³

³²We continue to ignore the fact that numerals which assign GEN in the subject position in Polish are themselves in the accusative (cf. fn. 21), from which certain subject–verb agreement facts follow. Nevertheless, this is the reason for different forms of the M1 numerals in (38) (*dwóch*.ACC=GEN) and (39c) (*dwaj*.NOM).

³³In the case of the non-flagging probe in (41), SAT:φ would also work here, or even a complete lack of probe, and it is not clear how to choose between these options. If NumP could be a maximal nominal projection in some environment, to the exclusion of DP or KP layers, then this insatiability would be needed to flag adjectival modifiers with an appropriate number and gender, but it is not clear to us whether NumP can ever be such a maximal extended projection. Moreover, while the N-selecting paucals behave like adjectives, they are still categorially numerals (and not, say, adjectives), as determined by certain category tests (especially the test of the distributive preposition *po* ‘each’, which combines with locative nominal phrases, but accusative numeral phrases; Przepiórkowski 2006).

- (40) Polish higher numerals: [INT:ϕ, SAT:–, FLAG:GEN], SEL:N
 (41) Polish paucal numerals: [INT:ϕ, SAT:–, FLAG:GEN], SEL:N.M1
 [INT:ϕ, SAT:–, FLAG:–], SEL:N

That is, Polish paucals are in principle ambiguous: they may have the usual GEN-flagging probe, but only when they select for M1 Ns, or they may have a non-GEN-flagging probe, without any gender restrictions.³⁴ Note that the first probe is the same as in the case of higher numerals (only selectional restrictions are slightly different), while the second differs from it minimally, in giving up GEN flagging (but selectional restrictions remain the same).

As expected, the difference between these two probes becomes invisible in oblique cases, e.g., DAT: whether the M1 noun is assigned GEN first or not, DAT is exponed – either as the more marked case than GEN (when the first probe in (41) is at play), or as the only case (in the case of the second probe):

- (42) przeciw dwóm {ochroniarzom / *ochroniarzy}
 against_{DAT} two.DAT guard.M1.PL. {DAT/*GEN}
 ‘against 2 (male) guards’ (POL)

In summary, Polish paucals do not require any substantial extensions to either the general approach to concord advocated here, or the particular analysis of Polish NNCs in terms of the *Expone Most Marked* rule.

5.2 Paucals in Russian

The syntax of Russian paucal constructions also depends on gender, but there is an additional complexity: when the paucal phrase occurs in a nominative or accusative position, most ad-paucal nouns occur in what looks like *singular* genitive, e.g., *stula* in (43),³⁵ instead of the expected plural genitive (as in the case of higher numerals in Polish and Russian or M1 paucal numerals in Polish) or plural nominative (as in the case of non-M1 paucal numerals in Polish),

- (43) dva {stula / rjadá / vyxodnyx}
 two.M.NOM/ACC chair.M.SG.GEN(?) row.M.SG?.GEN? exiting.M.PL.GEN(?)
 ‘2 {chairs / rows / days off}’ (RUS)

Assuming the usual repertoire of Russian cases and numbers, the form *stula* ‘chair’ may only be analyzed as SG.GEN.

However, there are 5 masculine nouns that may occur ad-paucally in a form that is not witnessed anywhere else: *rjad* ‘row’, *sled* ‘trace’, *čas* ‘hour’, *šag* ‘step’ and *šar* ‘sphere, ball’. When in an ad-paucal position, these nouns often display a stress pattern that is different from the SG.GEN stress, e.g., *rjadá* (with end stress) in (43) instead of the SG.GEN *rjáda* (with stem stress).³⁶

Apart from these 5 nouns, another problem for the generalization that Russian paucal numerals combine with SG.GEN complements is the fact that de-adjectival nouns, such as *vyxodnyx* ‘days off’ (lit. ‘exiting’) in (43), occur in what looks like PL.GEN. Moreover, adjectives modifying the supposedly SG.GEN nouns have PL.GEN forms, as in (44), in an apparent violation of concord.

- (44) dva krasnyx stula
 two.M.NOM/ACC red.PL.GEN(?) chair.M.SG.GEN(?)
 ‘2 red chairs’ (RUS)

While the examples above involve masculine nouns, neuter nouns behave similarly, i.e., even though they seem to be SG.GEN ad-paucally, they are modified by apparently PL.GEN adjectives in contemporary Russian. By contrast, feminine apparently SG.GEN nouns occur ad-paucally most often with plural adjectives syncretic between NOM and ACC, but sometimes also with GEN adjectives, e.g. (# marks the much rarer option):

- (45) dve {krasnye / #krasnyx} krovati
 two.F.NOM/ACC red.PL.NOM/ACC(?) / #PL.GEN(?) bed.F.SG.GEN(?)=PL.NOM/ACC
 ‘2 red beds’ (RUS)

³⁴While positing ambiguity is always suspicious, we believe that it is real and reflects diachronic development of paucal numerals from purely adjectival forms. The process of numeralization created ambiguities also at the other end of the scale, where words such as *tysiąc* ‘thousand’ and *milion* ‘million’ are ambiguous between numerals and nouns in contemporary Polish (see, e.g., Przepiórkowski 1999: 194–196 and references therein).

³⁵The full paradigm of *dva stula* ‘two chairs’ is given in the table in (47) below.

³⁶See Nesset and Nordrum 2019: 91 for corpus evidence that most of these nouns display the unusual stress ad-paucally in around 80% of text occurrences, with the exception of *šag* ‘step’, which has this non-SG.GEN stress in around 40% of occurrences.

Various sources give different generalizations about the co-occurrence of adjectives and nouns in ad-paucal positions. This is probably because, in such positions, both PL.NOM/ACC and PL.GEN adjectives could occur with apparently SG.GEN nouns bearing different genders to various degrees at various stages of Modern Russian. The detailed diachronic corpus investigation of this matter in Nessel 2020 demonstrates that the percentage of PL.GEN adjectival forms modifying masculine and neuter ad-paucal nouns grew steadily from around 15% in the mid-19th century to 99% in the early 21st century. In the case of feminine nouns, the reverse trend was observed: the percentage of PL.GEN forms diminished from around 80% at the end of the 19th century to 13% in the early 21st century. Hence, the contemporary generalization is that 1) only PL.GEN adjectives may occur with masculine or neuter nouns, as in (44), while 2) both PL.NOM/ACC and PL.GEN adjectives may occur with feminine nouns, but with a strong and growing preference for PL.NOM/ACC, as indicated in (45).

Stress also complicates the SG.GEN analysis of feminine nouns. While many feminine nouns have syncretic SG.GEN and PL.NOM/ACC forms (e.g., *krovati* in (45)), some distinguish these two possibilities by stress pattern, e.g., *storóný* ‘side.SG.GEN’ vs. *stórony* ‘side.PL.NOM/ACC’. In such cases, usually the apparently SG.GEN form occurs ad-paucally, regardless of the form of the accompanying adjective, but the PL.NOM/ACC stress pattern is also occasionally observed, especially in the case of *stórony* ‘side.PL.NOM/ACC’ (Nessel and Nordrum 2019).

Finally, as frequently noted (Franks 1994, 1995, Pesetsky 2013, Pereltsvaig 2010, Nessel and Nordrum 2019), also de-adjectival feminine nouns occur ad-paucally in what looks like plural rather than singular, e.g.:

- (46) tri {masterskie / masterskix / Ivanovy / Ivanovyx}
 three.F.NOM/ACC workshops.F.PL.{NOM/ACC / GEN}(?) Ivanova.F.PL.{NOM/ACC / GEN}(?)
 ‘3 {workshops / Ivanovas}’ (where *Ivanova* is an inflectionally adjectival surname) (RUS)

For all these reasons, contemporary analyses of Russian paucal constructions assuming that the noun is indeed SG.GEN are very rare (but see Nessel and Nordrum 2019). Instead, most analyses posit that ad-paucal nominals have a feature which is not observed elsewhere in Russian. The question is whether this unique feature is a special grammatical case, or a special grammatical number (or perhaps both; cf. Pereltsvaig 2010). While “paucal case” is more immediately compatible with the account so far (see §5.2.1), we will argue for “paucal number”, even though it requires certain extensions of this account (see §5.2.2).

Works advocating either “paucal case” or “paucal number” are sometimes less explicit about the value of the other category in paucal constructions. For example, if the case of *stula* ‘chair’ in *dva stula* ‘2 chairs’ is paucal, what is its number – singular or plural? Similarly, if *stula* has paucal number, what is its case in NOM/ACC positions? Below, we will assume values of the other categories that make the resulting analyses maximally similar to the standard analysis of higher numerals. This means that, on the “paucal case” hypothesis, the number of *stula* is plural in NOM/ACC, as it is in other case positions and in the case of higher numeral constructions; see the initial columns of the table in (47).³⁷ Similarly, on the “paucal number” hypothesis, the case of *stula* in NOM/ACC positions is taken to be GEN, as in the case of higher numerals; see the final columns in (47).³⁸

- (47) Values of number and case on the two paucal hypotheses:³⁹

position	“paucal case”		“paucal number”	
	‘two’	‘chairs’	‘two’	‘chairs’
NOM	<i>dva.NOM</i>	<i>stula.PL.PCL</i>	<i>dva.NOM</i>	<i>stula.PC.GEN</i>
ACC	<i>dva.ACC</i>	<i>stula.PL.PCL</i>	<i>dva.ACC</i>	<i>stula.PC.GEN</i>
GEN	<i>dvux.GEN</i>	<i>stul’ev.PL.GEN</i>	<i>dvux.GEN</i>	<i>stul’ev.PL.GEN</i>
DAT	<i>dvum.DAT</i>	<i>stuljam.PL.DAT</i>	<i>dvum.DAT</i>	<i>stuljam.PL.DAT</i>
LOC	<i>dvux.LOC</i>	<i>stuljax.PL.LOC</i>	<i>dvux.LOC</i>	<i>stuljax.PL.LOC</i>
INS	<i>dvumja.INS</i>	<i>stuljami.PL.INS</i>	<i>dvumja.INS</i>	<i>stuljami.PL.INS</i>

³⁷This is the explicit position of Ionin and Matushansky 2018: §6.3.6.

³⁸This is the explicit position of Asinari 2019, but see also references therein for alternatives and §5.3 for discussion.

³⁹In keeping with the convention of abbreviating case values with 3 letters and number values with 2 letters, and to decrease ambiguity, we abbreviate paucal *case* to PCL and paucal *number* to PC.

5.2.1 Paucal Case

The “paucal case” hypothesis has a long tradition in Russian linguistics (see, e.g., Zaliznjak 1967: 46–48), but it is also found in some more contemporary generative accounts (e.g., Franks 1995, Yadroff 1999, Rappaport 2002, Ionin and Matushansky 2018).⁴⁰ A formalization of this hypothesis in terms of the analyses of previous sections is immediate. The only difference between the higher numerals and the paucal numerals is the case value with which they flag their complements: GEN is the case associated with higher numerals, PCL is associated with paucal numerals:

(48) Russian higher numerals: [INT:ϕ, SAT:–, FLAG:GEN]

(49) Russian paucal numerals: [INT:ϕ, SAT:–, FLAG:PCL] (“paucal case” hypothesis)

On this hypothesis, and assuming case stacking and the *Exponere Most Marked* rule as in Polish, there are two case values stacked on ad-paucal nouns: PCL and the case specific to a given syntactic position. When that second case is NOM or ACC, the more marked PCL is exponed, but when the second case is an oblique (GEN, DAT, LOC, INS), that more marked oblique is exponed. Essentially this analysis is proposed in Ionin and Matushansky 2018: §§6.2–6.3.

This reasoning crucially depends on PCL being situated between ACC and GEN in an extended case hierarchy, as shown in (51):

(50) NOM < ACC < GEN < DAT < INS < ... (Caha 2009; repeated from (25))

(51) NOM < ACC < **PCL** < GEN < DAT < INS < ... (extended case hierarchy)

Fortunately for the “paucal case” hypothesis, case syncretism provides arguments for exactly this position of PCL.

Recall that, according to Caha 2009, only neighboring cases may display systematic syncretisms. Given that Russian attributive adjectives agree in case with the nouns they modify, not only ad-paucal nouns, but also ad-paucal adjectives must be assumed to bear PCL. For instance, according to the “paucal case” hypothesis, examples (44)–(45) have morphosyntactic features as in (52)–(53).

(52) dva krasnyx stula
two.M.NOM/ACC red.M.PL.PCL(=PL.GEN) chair.M.PL.PCL(=SG.GEN)
‘2 red chairs’ (RUS)

(53) dve {krasnye / #krasnyx} krovati
two.F.NOM/ACC red.F.PL.PCL(= {PL.NOM/ACC / #PL.GEN}) bed.F.PL.PCL(=SG.GEN=PL.NOM/ACC)
‘2 red beds’ (RUS)

As illustrated in (52), PL.PCL forms modifying masculine (or neuter) nouns are always syncretic with PL.GEN, while, as illustrated in (53), PL.PCL forms modifying feminine nouns are normally syncretic with PL.NOM/ACC, but with forms syncretic with PL.GEN also encountered. Hence, PCL must be a direct neighbor of both GEN and NOM/ACC, as in (51).⁴¹

While the “paucal case” hypothesis is immediately compatible with the approach to concord and NNCs advocated above, it is far from clear that it is true. One conceptual problem is that, on this hypothesis, the robust generalization about Russian that there are no gender distinctions in the plural is lost. This is because – as shown in (52)–(53) – different forms of PL.PCL adjectives occur with masculine/neuter nouns (only those that are syncretic with PL.GEN) and with feminine nouns (mainly those that are syncretic with PL.NOM/ACC). The “paucal number” hypothesis is free from this problem, as demonstrated in the next subsection.

Another conceptual argument against “paucal case” is that, unlike the cross-linguistically widely attested paucal number, “paucal case” does not seem to occur anywhere beyond – hypothetically – East Slavic paucals.

Moreover, the “paucal number” hypothesis meshes better with the diachronic developments of Russian paucal constructions, which started with the gradual loss of the dual number during the Old Russian period; a reanalysis of the last vestiges of dual number as paucal number is less dramatic than a reanalysis of a vanishing number as a totally new case (not observed anywhere else, as mentioned above).

⁴⁰The Russian term *sčëtnyj padež*, lit. ‘count case’, has been variously translated as “numeral” case (e.g., Nessel and Nordrum 2019) or as “adnumeral”, “countative” or “calculative” case (various web pages). However, in most of the English-language generative literature it is simply called “paucal” case (e.g., Franks 1994, 1995, Rappaport 2002, Ionin and Matushansky 2018; cf. “adpaucal case” in Yadroff 1999: 146), so we also use this term here.

⁴¹Incidentally, no such (total) syncretisms of adjectival forms need to be postulated on the “paucal number” account below.

Finally, it is a robust cross-linguistic generalization that, unlike pronouns, lexical nouns do not exhibit case-based root suppletion and only show very limited stem allomorphy (limited to NOM and perhaps ACC, if it is syncretic with NOM); see, e.g., Moskal 2015, McFadden 2018, and Smith *et al.* 2019. However, on the “paucal case” hypothesis, case-based root suppletion would be observed which would distinguish PCL from all other cases in the plural:

- (54) a. {rebënok / rebënka / – / rebëнку / rebënke / rebëнком}
 child.SG.{NOM / ACC=GEN / ~~PCL~~ / DAT / LOC / INS}
 b. {deti / detej / **rebënka** / detjam / detjax / det'mi}
 children.PL.{NOM / ACC=GEN / **PCL** / DAT / LOC / INS}
 c. {dva rebënka / pjat' detej}
 two.NOM/ACC children.PL.PCL five.NOM/ACC children.PL.GEN
 ‘{two children / five children}’ (RUS)

As shown in (54a–b), the singular root for ‘child’ in Russian is *rebënk*, while the plural root is normally *det*. The single exception is the hypothetical PL.PCL form *rebënka*. This is not a matter of a special ad-numeral form, as shown in (54c): the ad-numeral plural genitive is the regular *detej*.⁴² Hence, the only explanation for this suppletion on the “paucal case” analysis is that it is case-driven, contrary to the generalization mentioned above. Additionally, this analysis would result in a clear violation of the robust *ABA generalization (on which see, e.g., Bobaljik 2012 and §6.4 below, esp., fn. 63 there), with the root in PL.PCL (the B in *ABA) different from the roots in surrounding cases in the plural (the As in *ABA).⁴³ By contrast, on the “paucal number” analysis below, this root suppletion is number-driven, with SG and PC forms based on *rebënk*, and PL forms based on *det*.⁴⁴

While perhaps none of these problems alone is necessarily fatal for the “paucal case” hypothesis, together they amply justify an attempt at substantiating the alternative “paucal number” hypothesis.

5.2.2 Paucal Number

The “paucal number” hypothesis seems to be more frequently assumed in western than in Russian linguistics (e.g., Rakhlin 2003, Bailyn and Nevins 2008, Asinari 2019, Breu 2020; cf. Corbett 1993: 14–16).⁴⁵ It is more attractive cross-linguistically, given that paucal number – but not paucal case – is witnessed in other languages, and it is perhaps more compatible with diachrony, given that the development of the paucal construction was triggered by the loss of dual number in Old Russian (Neset 2020), but – as discussed below – it faces the conceptual problem that, unlike in most languages with paucal number, the apparently paucal forms of Russian nouns cannot alone, without paucal numerals, express semantic paucal number.

On the analysis proposed in this subsection, the expounded features of the two running examples (44)–(45) are as indicated in (55)–(56).

- (55) dva krasnyx stula
 two.M.NOM/ACC red.PL.GEN chair.M.PC.GEN(=SG.GEN)
 ‘2 red chairs’ (RUS)

⁴²Such a special ad-numeral plural form is witnessed in the case of *čelovek* ‘person.SG’, which in the plural is normally based on the root *ljud* (e.g., *ljudi* ‘people.PL.NOM’), but in ad-numeral contexts, in NOM, ACC, and GEN positions, it reverts back to the singular root *čelovek* (e.g., *dva čeloveka* ‘two people’, *pjat’ čelovek* ‘five people’). On the current analysis, this suppletion would be conditioned by the feature NUM(eral), rather than by case (with GEN contexts requiring some additional explanation). Similarly, *god* ‘year’ has ad-numeral forms showing the root suppletion *let* (normally a PL.GEN form of *leto* ‘summer’), but only in plural, not in paucal (e.g., *odin god* ‘one year.SG.NOM’, *dva goda* ‘two years.PC.GEN’, but *pjat’ let* ‘five years.PL.GEN’). On the current analysis, this form would expone the feature bundle containing [NUM, PL], i.e., it would be conditioned on category (resulting from flagging by the numeral) and number – not on case. Compare Ionin and Matushansky 2018: §6.6.2, from which the suppletion data discussed here are drawn, as well as Pancheva 2018, Franks 2018, and Ionin and Matushansky 2018: §6.6.3 on ad-numeral forms in Bulgarian (a language with overt case limited to pronouns).

⁴³*ABA is violated in (54b) not only in terms of root suppletion, but also in terms of syncretism, given that (ACC.PL = GEN.PL) ≠ PCL.PL, and that the ACC < PCL < GEN part of the case hierarchy in (51) is crucial for the “paucal case” explanation of case patterns in Russian NNCs.

⁴⁴See fn. 52 on the possibility that, in Russian, PL is more marked than SG and PC. Moreover, given that such feature hierarchies may be partially rather than linearly ordered (see, e.g., Zompi 2023 on person and §6.4 below and references there on case), another possibility is that both PC and PL are more marked than SG, with no order between PC and PL. On either analysis of Russian number, our proposal that suppletion may target PL to the exception of SG and PC does not violate the *ABA generalization.

⁴⁵See also Pesetsky 2013 for a different analysis in terms of exceptional number.

- (56) dve {krasnye / #krasnyx} krovati
 two.F.NOM/ACC red.PL.{NOM/ACC / #GEN} bed.F.PC.GEN(=SG.GEN=PL.NOM/ACC)
 ‘2 red beds’ (RUS)

Note that, on this analysis, the forms of adjectives are what they seem: plural genitive (rather than paucal genitive or plural paucal) in (55) (and in (56)) and PL.NOM/ACC (rather than PC.NOM/ACC or PL.PCL) in (56).

The main idea of the analysis of Russian paucals offered below is this:

- paucal numerals – and only they – enter the derivation lexically specified for the paucal number,⁴⁶
- nouns in quantity contexts enter the derivation specified for the plural;
- as a result of the numeral–noun Agree, each ϕ -feature bundle is copied to the other item;
- this way nouns end up with the paucal number, but only in ad-paucal positions.

This idea immediately accounts for the conceptual problem with the “paucal number” hypothesis mentioned above: if Russian has paucal number, why cannot paucal forms of nouns be used alone to express paucal quantity (Corbett 1993, 2000: 22–23, fn. 18)? For example, why does *vladelec stula* ‘owner.NOM chair.GEN.PC=SG’ mean only ‘owner of a chair’ and why can it not be used to express ‘owner of a couple of chairs’? The answer is a lexical idiosyncrasy grounded in diachrony: the only lexical items in contemporary Russian that bear the paucal number are the paucal numerals, born on the ashes of dual (Neset 2020). Nouns may receive paucal number only when they are probed by such paucal numerals, so only then can they express paucal quantity. Hence, the analysis offered here is free from the main conceptual problem with the “paucal number” approach.

In order to substantiate this analysis, let us first recall that, as a separate grammatical category, Slavic numerals stem from adjectives (paucals) and nouns (higher numerals), and share certain features with both these more established categories. Words expressing smaller numbers display more adjectival behavior and those expressing higher numbers have more features common with nouns (Corbett 1978), to the extent that *odin* ‘one’ is a pure adjective (in particular, it inflects for number, unlike true numerals, e.g., *odni štany* ‘one pair of trousers’, lit. ‘one.PL trousers.PL’) and *million* ‘million’ is sometimes claimed to be a pure noun (in particular, it has a specific – masculine – gender and it takes a genitive noun even in oblique cases, both unlike true numerals).⁴⁷

Given that Slavic numerals stem from adjectives and nouns, we posit that their featural makeup is somewhere between that of adjectives and that of nouns. A conspicuous difference between these two categories in north Slavic is that adjectives do not have any lexically specified values of ϕ features, while nouns are lexically inserted with an idiosyncratic gender and sometimes idiosyncratic number, the latter conspicuous in *plurale tantum* and *singulare tantum* nouns (e.g., *štany* ‘trousers.PL’ and *kto* ‘who.SG’). We claim that, by comparison, north Slavic numerals are an intermediate class: they have lexically valued number, but unvalued gender. Additionally, most numerals – including some paucals – are more like nouns in combining with GEN complements, while some paucals are more like adjectives in the lack of GEN-flagging. However, even when numerals GEN-flag their NP complements, this genitive is overridden by more marked oblique cases, unlike in the case of nouns, whose GEN complements are KPs impenetrable to further case flagging. This is summarized in (57).

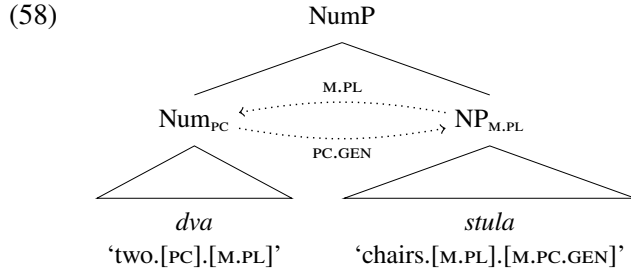
(57)	Adjectives (incl. <i>odin</i> ‘1’)	N u m e r a l s			N o u n s (incl. <i>million</i> ‘10 ⁶ ’)
		P a u c a l s	High		
		adjectivish	nounish		
lexical number	–	+	+	+	(+)
GEN complements. . .	–	–	+	+	+
. . . also in oblique	–	–	–	–	+
lexical gender	–	–	–	–	+

⁴⁶We again assume here for simplicity that items in the lexical array may be specified for features such as gender and number. Nothing seems to hinge on this assumption and the analysis offered here is compatible with views on which all grammatical features are introduced by functional heads. For example, the statement that in Russian only and all paucal numerals are lexically specified as bearing the paucal number may be translated into the property of the Number/# functional head that it may bear the paucal feature only in the context of roots of paucal numerals.

⁴⁷In fact, Russian *million* may be ambiguous between a noun and a numeral (Ionin and Matushansky 2018: §6.2.6), similarly to Polish *milion* ‘million’ (cf. fn. 34).

As suggested above, what is special about Russian paucals is that their lexically specified number is, well, paucal (diachronically stemming from the old dual). Specifically, we claim that paucal numerals are the only lexical items specified for PC in contemporary Russian; in particular, higher numerals are specified as plural (similarly to all numerals in Polish) and nouns enter the derivation as singular or plural.

Given that Russian countable nouns are typically plural in quantity environments, regardless of the number of the selector (e.g., *para stul'ev* ‘a pair of chairs’, lit. ‘pair.F.SG.NOM chairs.M.PL.GEN’), the first step in the construction of a simple paucal phrase, *dva stula* ‘two chairs’, is as in (58).



As explained above, the numeral enters the derivation specified as paucal, [PC], and the noun as plural, here [M.PL]. There are two results of Agree between them: 1) copying of the noun’s ϕ features to the numeral, and 2) flagging the noun with GEN, but also – as in the system of Clem and Deal 2026 – with the numeral’s ϕ features.⁴⁸ Hence, each item has two feature bundles, as indicated on the terminals in (58).⁴⁹ These feature bundles form a stack; this is a natural generalization not only of the analysis of NNCs in §4 in terms of case stacks, but also of the analysis of Grishin 2024, on which ϕ features agglomerated as a result of multiple Agree form a stack, and the analysis of Clem and Deal 2026, on which feature bundles combine different kinds of features (at least categorial and ϕ). While all stack elements are visible to morphological exponence rules, we assume together with previous literature that syntactic processes only see the top of the stack.⁵⁰ For example, once the numeral agrees with the nominal as in (58), the feature bundle on Num (and, hence, also on NumP) active in ensuing syntactic processes will be [M.PL], which is on top of the stack, and not the original [PC] at the bottom of the stack.

Before we move to the next step, let us explicitly introduce an assumption responsible for the strengthening of [PC.GEN] to [M.PC.GEN] in *stula* ‘chairs.[M.PL].[M.PC.GEN]’ in the tree above:

(59) Priority Union of Bundles on Stack (PUBS)

When a feature bundle is added to a stack, it undergoes priority union with the bundle on top of this stack. The name of this principle refers to an operation on syntactic and semantic feature structures assumed within Lexical Functional Grammar (see, e.g., Kaplan 1987), but also used elsewhere; for example, it has recently been considered “as a candidate for inclusion in phonological Universal Grammar” (Reiss 2022: 199).⁵¹ For the purpose of this analysis, Priority Union may be defined as in (60).

(60) The *Priority Union* B/A of two bundles of features A and B is an extension of B with those features of A which do not contradict those in B.

For example, the priority union of B = [PL.GEN] (the new feature bundle) with A = [M.PL] (the top of the stack) is B/A = [M.PL.GEN]. This is because gender information is missing in B and adding it does not introduce any inconsistency. This is the reason why the top of the stack in *stula* ‘chairs.[M.PL].[M.PC.GEN]’ in (58) is [M.PC.GEN] rather than simply [PC.GEN]. On the other hand, the priority union of B = [M.PL] with A = [PC] is B/A = [M.PL] = B, as adding PC to

⁴⁸We ignore the possibility of category flagging, as it does not seem to play any role here, but it may be crucial to explain some ad-numeral forms, e.g., the suppletive *let* lit. ‘summers.PL.GEN’ used to expone the ad-numeral ‘year.PL.GEN’ instead of the regular *godov* ‘year.PL.GEN’.

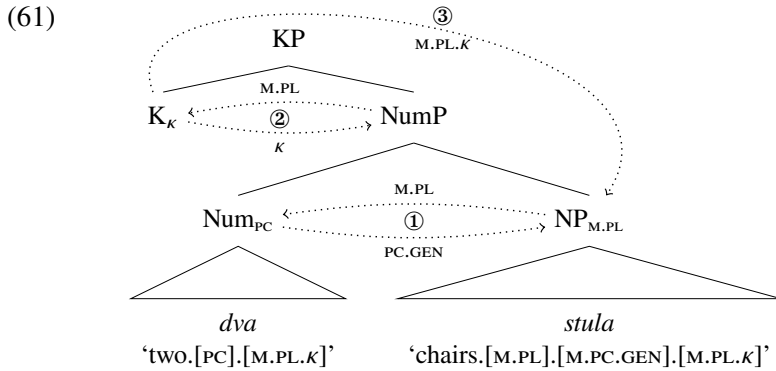
⁴⁹The presence of M in [M.PC.GEN] on the right terminal is explained immediately below. As before, subscripts on pre-terminals (here, on Num and N) indicate the inherent (lexical) features of the relevant items (but cf. fn. 46).

⁵⁰See, e.g., Bruening and Al Khalaf 2020 and Bruening 2023 for analyses employing stacks and assuming that certain processes only see the top of a stack (but also see Patejuk and Przepiórkowski 2023 and Przepiórkowski and Patejuk 2025 for a critique of these particular analyses).

⁵¹See also Müller 2007 on a related enrichment mechanism – dual to impoverishment in Distributed Morphology – proposed to account for multiple exponence of features, e.g., number in German.

[M.PL] would introduce inconsistency.

We are now ready to move to the next step, Merge of NumP with K:



After the initial numeral–noun Agree step discussed above, the ϕ features at the top of Num(P)’s stack are [M.PL], so it is this bundle that gets copied to K from NumP in the second step, when the NumP is flagged with κ . Hence, subsequent flagging by K involves the [M.PL. κ] bundle, resulting in the presence of this bundle at the top of N(P)’s stack. If κ is NOM or ACC, the [M.PC.GEN] bundle is more marked, so the paucal number is exponed on the genitive noun. By contrast, when κ is DAT, LOC, or INS, the outermost bundle involving κ is the one that is most marked, so it is exponed, with the appropriate case and ordinary plural number (originally introduced by the noun itself!), as shown in the table in (47) and, for DAT, in (62) below.

- (62) dvum {stuljam / *stula}
 two.DAT chair.M.{PL.DAT / *PC.GEN(=SG.GEN)}
 ‘2 red chairs’ (RUS)

Interestingly, when κ is GEN and, hence, the two cased feature bundles on the noun are the paucal [M.PC.GEN] and, at the top of the stack, the plural [M.PL.GEN], it is the outermost bundle that is exponed – the noun has the plural genitive form, not the paucal genitive form:

- (63) dvux {stul’ev / *stula}
 two.GEN chair.M.{PL.GEN / *PC.GEN(=SG.GEN)}
 ‘2 red chairs’ (RUS)

There are various ways to ensure this result.⁵² The analysis that we adopt here for concreteness is that markedness is determined solely by case, and when there is a draw, the outermost of the maximally marked bundles is exponed. That is, we replace the *Expone Most Marked* rule in (23) with the *Expone Outer Most Marked* rule in (64).⁵³

- (64) *Expone Outer Most Marked*: Expone only the outermost of maximally marked feature bundles.

Note that, when there is a single bundle that is most marked, as in all cases of feature bundle competition considered in previous sections, (64) has the same effect as the *Expone Most Marked* rule in (23) assumed so far.

The analysis above does not depend on the gender of the noun, but such a dependence is needed when adjectival modifiers are considered. It is at this point that we need to be more specific about the nature of the probe on Num.

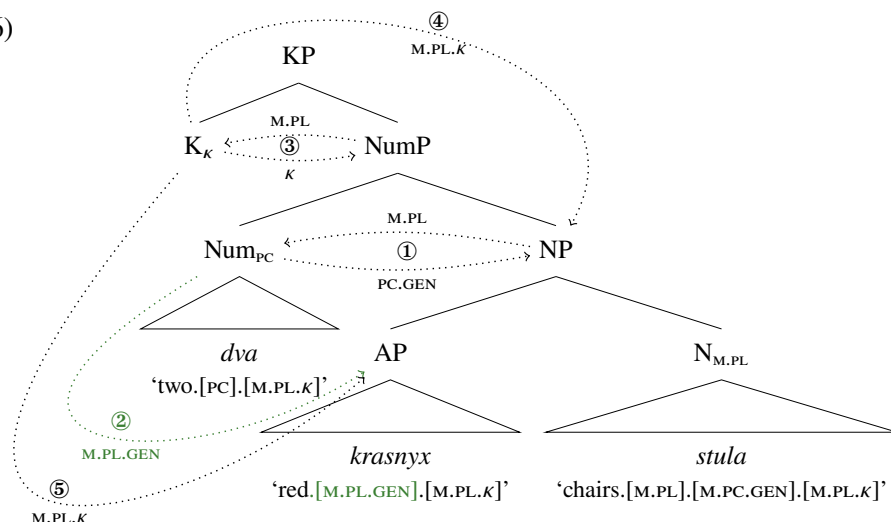
Consider the tree in (66) for the paucal phrase in (55), repeated below as (65), which differs from the previous tree in (61) in the presence of an adjectival modifier.

- (65) dva krasnyx stula
 two.M.NOM/ACC red.PL.GEN chair.M.PC.GEN(=SG.GEN)
 ‘2 red chairs’ (RUS)

⁵²One would be to assume a markedness metric on which, at least in Russian, plural is more marked than paucal and, hence, the feature bundle [M.PL.GEN] is more marked than [M.PC.GEN]. This seems to go against the commonly assumed hierarchy SG < PL < DU/PC (see, e.g., Nevins 2011 and references therein), but – as noted in Smith *et al.* 2019 – there exist languages where suppletion facts suggest that dual is *less* marked than plural, one of them being Slovenian. So, given that both Slovenian dual and Russian paucal stem from Proto-Slavic dual, the possibility that Russian plural is more marked than paucal should not be rejected without investigation.

⁵³The name of this rule is the result of *of*-deletion and haplology of *most* in *Expone Outermost of Most Marked*.

(66)



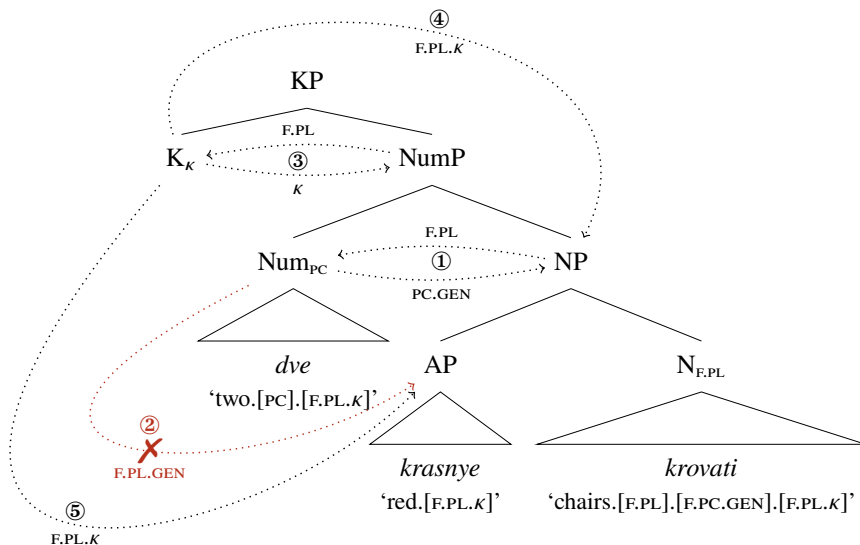
Steps marked ①, ③, and ④ are exactly the three steps considered above. Additionally, given that the probe on K is insatiable, it also flags the AP with [M.PL.κ], as discussed in §3 (see step ⑤). The crucial question is whether the probe on Num should also be insatiable, as argued for higher numerals in §4.2.3 and assumed in (66) (see step ② and the related bits in green), or not.

If the probe on Num is insatiable, it does not stop after Agree with the NP, but it also flags the AP with [M.PL.GEN] (as, after step ①, M.PL is on top of Num(P)'s stack), so this feature bundle is added to the A(P)'s stack in step ②, before [M.PL.κ] is added in step ⑤. As explained in §4.2.3, this predicts that the AP is exponed as PL.GEN in NOM/ACC positions, given that [M.PL.GEN] is more marked than [M.PL.κ] then, and as [M.PL.κ] in other positions. This is the right prediction for all masculine and neuter nouns (as illustrated in the table in (47)), and it seems to still be a possibility for feminine nouns.

But what about the more common situation in feminine paucal constructions, where APs modifying the F noun are in NOM/ACC? It turns out that this possibility is readily accommodated by the system proposed here, on the assumption that in such cases the probe is immediately satiable. Then step ② does not take place, and the result is as in the tree in (68) for the paucal phrase in (56), partially repeated in (67).

- (67) dve krasnye krovati
 two.F.NOM/ACC red.PL.NOM/ACC bed.F.PC.NOM(=SG.GEN=PL.NOM/ACC)
 '2 red beds' (RUS)

(68)



Note that the only feature bundle on the A(P) is now [F.PL.K], so such modifiers are truly NOM/ACC in a NOM/ACC environment, as illustrated in (67)–(68), even though the feminine noun is itself expounded as [F.PC.GEN] – usually the same as SG.GEN rather than PL.NOM, where suprasegmental features make this distinction visible.

- (69) Russian/Polish higher numerals: [INT: ϕ , SAT:-, FLAG:GEN], SEL:N
(70) Russian paucal numerals: [INT: ϕ , SAT:-, FLAG:GEN], SEL:N
[INT: ϕ , SAT: ϕ , FLAG:GEN], SEL:N.F
(71) Polish paucal numerals: [INT: ϕ , SAT:-, FLAG:GEN], SEL:N.M1
[INT: ϕ , SAT:-, FLAG:-], SEL:N

5.3 Interim Summary and Comparisons

While concentrating on the “paucal number” hypothesis, we noted that the “paucal case” approach is also immediately formalizable within the concord framework advocated in this paper. Nevertheless, we pointed out various problems with “paucal case”, which provided impetus for developing the “paucal number” analysis.

- (72) {dolžna / dolžen} Maše bol'suju summu
 due.A.F/M Masha.DAT large.ACC sum.ACC
 'owing Masha a large sum' (RUS)

⁵⁴This suggestion appears also in Corbett 1993: 15.

(animate) ACC. In short, we are not aware of any convincing arguments against the analysis of paucal constructions in §5.2.2, on which the noun bears the genitive in NOM/ACC positions, just as in other numeral constructions.

Let us finally note that, while – as mentioned above – we share with Pesetsky 2013 the analysis of nouns in NOM/ACC paucal constructions as truly genitive, we do not agree with another crucial aspect of that account. For Pesetsky (2013), such nouns do not bear paucal number, but are actually numberless; what is usually taken to be a paucal *numeral* is actually a grammatical *number morpheme*. This, however, not only reintroduces the dual number (given the presence of the number morpheme *dva* ‘two’), generally considered long lost in Russian, but also introduces the cross-linguistically rare trial (*tri* ‘three’), the highly questionable (Corbett 2000: §2.2.5) quadral (*četyre* ‘four’), as well as hitherto unknown grammatical numbers corresponding to *poltora* ‘one and a half’, and also to *pol-* ‘half’ and *četvert’* ‘quarter’.⁵⁵ Instead of introducing all these number values into Russian grammar, some of which have not been attested anywhere else so far, we posit that Russian simply has some vestiges of paucal number, which is cross-linguistically well-attested (Corbett 2000: §2.2.4).

6 Heterogeneous Patterns in Finnic

Norris (2018) claims that Estonian pseudopartitives are a challenge for analyses of heterogeneous patterns based on case stacking. In this section, we first present the relevant Estonian data (in §6.1) and the challenge for case-stacking accounts (in §6.2). We show (in §6.3) that while the relevant data are problematic for an account based on the *Expone Outermost* rule, they are not problematic for our version of exponence rules, based on markedness. We motivate (in §6.4) our modification of Caha’s (2009) case hierarchy needed to account for Estonian pseudopartitives and – after comparison with Norris 2018 (in §6.5) – we finish (in §6.6) with some cross-linguistic considerations, including a brief comparison to another Finnic language, Finnish.

6.1 Estonian Pseudopartitives

Pseudopartitive constructions contain two nouns, the first of which (N1) indicates the quantity of the entity or entities denoted by the second (N2), as in English *piece of bread* or *bag of potatoes*. In Estonian, the two nouns bear the same case in most syntactic positions, with the exception of nominative and accusative, as illustrated with the Estonian for ‘piece of bread’ in (73).

(73)	position	‘piece.sg’	‘bread.sg’
	NOM	tükk. NOM	leiba. PAR
	ACC	tüki. ACC	leiba. PAR
	PAR	tükki. PAR	leiba. PAR
	GEN	tüki. GEN	leiva. GEN
	ILL	tüki-sse. ILL	leiva-sse. ILL
	INE	tüki-s. INE	leiva-s. INE
	ELA	tüki-st. ELA	leiva-st. ELA
	ALL	tüki-le. ALL	leiva-le. ALL
	ADE	tüki-l. ADE	leiva-l. ADE
	ABL	tüki-lt. ABL	leiva-lt. ABL
	TRL	tüki-ks. TRL	leiva-ks. TRL
	TER	tüki. GEN	leiva. GEN-ni.TER
	ESS	tüki. GEN	leiva. GEN-na.ESS
	ABE	tüki. GEN	leiva. GEN-ta.ABE
	COM	tüki. GEN	leiva. GEN-ga.COM

Two aspects of this table do not follow tradition. First, Norris (2014, 2018) convincingly argues that four of the traditional Estonian cases – terminative, essive, abessive, and comitative (shown at the bottom of the table) – are actually clitic postpositions combining with a nominal phrase in the genitive. That is why these “cases” are only

⁵⁵See Ionin and Matushansky 2018: §§6.3.7 and 6.4.3 on the paucal status of these two words.

marked on the last word of the phrase, N2, rather than on both N1 and N2.⁵⁶ Second, the Estonian accusative is not usually recognized in traditional grammars, as it is not autonomous: it is always syncretic either with genitive (in the singular, as in (73)) or with nominative (in the plural). In this respect, Estonian is very similar to Finnish, which, however, has some autonomous (i.e., non-syncretic) accusative forms of pronouns. In Estonian, there is even less evidence for accusative as a separate case, but a convincing argument may be found in Norris 2014, 2018.⁵⁷

Note that the heterogeneous vs. homogeneous pattern observed here is very similar to that in Slavic numerals, but with two important differences. First, the non-agreeing case on N2 in the heterogeneous pattern is partitive rather than genitive, as in Slavic.⁵⁸ Second, the form of N1 in the accusative (here, *tüki*) is not the same as in the nominative (here, *tükk*), while Slavic numerals usually have the same NOM and ACC forms. This second difference is claimed in Norris 2018 to be incompatible with case-stacking approaches.

6.2 Challenge for Case Stacking?

Norris (2018) considers two variants of the case-stacking approach, differing in the exponence rules:

(74) *Pronounce Outermost*: Only the outermost case in a case stack is realized.

(75) *Pronounce Outermost Overt*: Pronounce the outermost case with an overt realization.

The rule in (74) is a paraphrase of Pesetsky's (2013) *One-Suffix Rule*, which we generalized to the *Exponence Outermost* rule in (22), repeated below as (76).

(76) *Exponence Outermost*: Exponence only the outermost feature bundle. (cf. Pesetsky 2013: 11, (7))

A rule like (74) or (76) suffices in Pesetsky 2013 to exponence the genitive on the noun within numeral–noun constructions in nominative and accusative positions, because various special mechanisms make sure that NOM and ACC are never assigned to the noun within the NNC.⁵⁹ As a result, the case stack of the noun in NNC only contains GEN, so this case is exponenced.

However, this analysis does not extend to Estonian pseudopartitives, as nominative and accusative have different forms (e.g., *tükk* vs. *tüki* in (73)), so it cannot be claimed that both reflect the lack of external case assignment; even if NOM is analyzed as lack of case, ACC must be analyzed as a specific case. Hence, after the pseudopartitive N1 flags N2 with PAR, the whole construction is flagged with ACC, so N2 should have two cases on its stack: the innermost PAR and the outermost ACC. But then ACC should be exponenced (e.g., as *tüki leiva* 'piece.ACC=GEN bread.ACC=GEN', which is grammatical in genitive positions but not in true accusative positions), rather than the actually observed PAR (e.g., *tüki leiba* 'piece.ACC bread.PAR', grammatical in accusative positions).

The version of the exponence rule in (75), cited in Norris 2018 as assumed in Baker and Vinokurova 2010 and Brattico 2008, 2010, 2011, also does not suffice here. Even if Estonian nominative is analyzed as a case without an overt realization – in which case (75) correctly predicts that NOM loses to PAR, even though PAR is deeper on the PAR.NOM stack – the same cannot be said about the accusative, so the ACC on the PAR.ACC stack should be exponenced in accusative positions, contrary to facts.

On the basis of this reasoning, Norris (2018: 559) concludes that Estonian pseudopartitives “pose challenges to analyses based on so-called case stacking or multiple case assignment to the quantified phrase”. We take up this challenge in the following subsections.

6.3 Case Stacking Analysis

Given Norris's (2018) assumptions about the internal structure of Estonian pseudopartitives, the analysis of the observable case patterns in terms of insatiable Agree and case stacking is almost immediate.⁶⁰

⁵⁶See Weisser 2020 for an analogous claim based on Estonian coordinate structures.

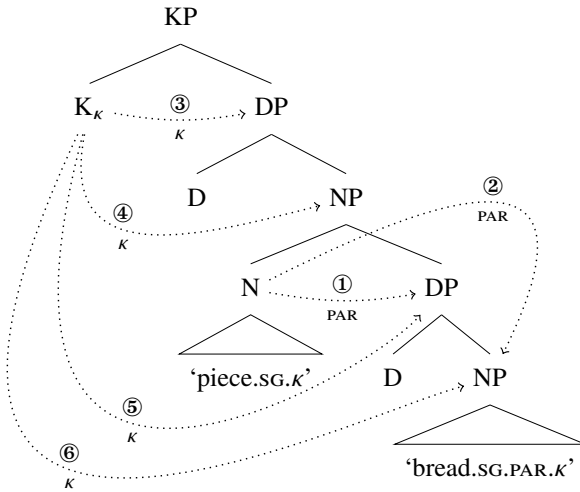
⁵⁷In brief, while N1 has the same form in accusative and genitive positions, N2 is partitive in true (ad-verbal) accusative positions, while it is genitive in true (ad-nominal and ad-prepositional) genitive positions.

⁵⁸In Russian, some nouns have a separate singular form that may be used in some partitive contexts, but the apparently SG.GEN form used in paucal constructions is always genitive, not partitive, e.g., *dva {čaja/*čaju}* 'two teas.PC.GEN=SG.{GEN/*PAR}'.

⁵⁹These special mechanisms include special properties of Determiners, which normally assign the nominative, but not to the noun within an NNC, and special properties of Verbs, which normally assign the accusative, but not to NNCs. (Cf. Przepiórkowski 2026.)

⁶⁰We follow Norris 2018 in concentrating on case concord, to the exclusion of ϕ concord. Given that N1 and N2 may have different grammatical numbers, some care needs to be taken to avoid flagging N2 with N1's number. One possibility is that, in Estonian, there are two

(77)



The shared assumption of the current account and Norris’s (2018) analysis of Estonian is that maximal nominal projections are KPs, i.e., locality boundaries (*phases* for Norris, here also *horizons*). Additionally, Norris (2018) assumes that Estonian has DPs, although Ds are usually phonetically empty. Importantly, what is special about Estonian pseudopartitive constructions according to Norris (2018) is that the N1 selects for a DP, rather than the usual KP. As DPs are not locality boundaries, their constituents are visible to external processes. We adopt these assumptions and note that, in terms of our analysis, this means that the probe on K in (77) flags with κ not only its sister DP and the NP this DP immediately dominates (see steps three and four), but also the lower DP and NP constituents (steps five and six).

However, unlike Norris (2018), we assume that, just as Slavic numerals flag accessible constituents with GEN, Estonian pseudopartitive nouns flag accessible constituents with PAR (see steps one and two in (77)). As this happens before K merges, the lower N2 has two cases on its stack, PAR at the bottom and κ at the top (while the higher N1 is only flagged with κ), as indicated in (77).⁶¹

Now, as correctly noted in Norris 2018, if a rule like *Expone Outermost* is assumed, then this analysis incorrectly predicts that the N2 is NOM or ACC in a NOM or ACC environment, while in fact it is PAR (recall table (73)). However, assuming our *Expone Outer Most Marked* rule (64), repeated below as (78), it is enough for PAR to be more marked than NOM and ACC and not more marked than all the other Estonian cases to derive the pattern in table (73).

(78) *Expone Outer Most Marked*: Expone only the outermost of maximally marked feature bundles.

Independent arguments for exactly this view of case markedness in Estonian are adduced in the following subsection.

6.4 Estonian Case Hierarchy

We share Norris’s (2018: 544) objective “to avoid a stipulated case hierarchy—that is, encoding a brute force statement like ‘genitive is stronger than partitive, accusative is weaker’ in the grammar”. So far, we have assumed Caha’s (2009) case hierarchy, which was not stipulated but motivated on the basis of numerous phenomena, including case syncretism and morphological containment. While partitive is not included in the original hierarchy repeated in (79), Caha (2009: 114–115) briefly considers the Estonian partitive and extends the hierarchy to (80).

(79) NOM < ACC < GEN < DAT < INS < ... (Caha 2009: 30; repeated from (25) and (50))

(80) NOM < ACC < **GEN** < **PAR** < DAT < INS < ... (cf. Caha 2009: 115)

This hierarchy is only partially compatible with our assumptions. Partitive is more marked than nominative and

separate probes on K: an insatiable probe that only flags constituents with κ , and an immediately satiable probe that Agrees with the sister DP in ϕ features, making these ϕ features present on the KP maximal nominal projection. If so, agreement in number between either of the two nouns and their corresponding modifiers would be the result of an insatiable probe on D (which would not be able to reach beyond the lower DP horizon).

⁶¹Technically, for this to work, the PAR-flagging probe on the higher N1 cannot have the default N horizon (as then no flagging would take place, given that the DP inherits the N feature; see Keine 2019: 37–40), but rather the wider K horizon (such “wider” horizons are part of the locality theory of Keine 2019, 2020 and they are argued there to be learnable).

accusative, accounting for the PAR expounded on N2 in NOM and ACC positions, but it is also supposedly more marked than genitive, predicting PAR on N2 also in GEN positions, contrary to facts: as shown in table (73), the N2 ‘bread’ is expounded as the GEN *leiva* rather than the PAR *leiba* in genitive positions.

The remainder of this subsection demonstrates that, independently of pseudopartitive constructions, Estonian morphological facts require a different extension of Caha’s case hierarchy in (79), namely, the extension in (81).

(81) NOM < ACC < {GEN § PAR} < obliques (Estonian)

On this extension, both PAR and GEN are more marked than NOM and ACC and both are less marked than oblique cases, whatever the exact order of markedness within such obliques; these relations are just as in Caha’s (80). However, unlike in (80), PAR and GEN are not comparable here: neither is more marked than the other.

Note that this hierarchy immediately explains the Estonian heterogeneous vs. homogeneous pattern in pseudopartitive constructions. In NOM and ACC positions, the case stacks on N2 are PAR.NOM and PAR.ACC respectively, so the more marked PAR is expounded. In oblique case positions such as illative or inessive, the oblique is more marked than PAR, so it is expounded. Crucially, in genitive positions, the case stack on N2 is PAR.GEN, and neither of these two cases is more marked than the other, so the *Expone Outer Most Marked* rule repeated below as (82) predicts that the outermost of these – the genitive – is expounded.

(82) *Expone Outer Most Marked*: Expone only the outermost of maximally marked feature bundles.

As illustrated in table (73), this prediction is correct. Hence, if this analysis is on the right track, it provides an independent argument for the *Expone Outer Most Marked* rule – motivated earlier only by the behavior of Russian paucals – as opposed to similar rules found in the literature (*Expone Outermost*, *Expone Most Marked*, etc.).

In order to show that this analysis is on the right track, we need to demonstrate – on the basis of independent morphological facts – that the right case hierarchy for Estonian is the partial order in (81) rather than the total order in (80). Let us first examine the reasoning that led Caha (2009) to the extension in (80). This reasoning is based on Estonian data in tables (83) and (84).⁶²

(83) Estonian structural cases, plural

position	‘flag.PL’	‘church.PL’	‘person.PL’
NOM	lipud	kirikud	inimesed
ACC	lipud	kirikud	inimesed
GEN	lippude	kirikute	inimeste
PAR	lippusid	kirikusid	inimesi

(84) Estonian structural cases, singular

position	‘corridor.SG’	‘person.SG’	‘year.SG’	‘church.SG’
NOM	koridor	inimene	aasta	kirik
ACC	koridor-i	inimes-e	aasta	kirik-u
GEN	koridor-i	inimes-e	aasta	kirik-u
PAR	koridor-i	inimes-t	aasta-t	kirik-u-t

Recall first that Estonian ACC is systematically syncretic with NOM in the plural. This is illustrated in (83). Given that both PAR.PL and GEN.PL may have forms different from NOM.PL=ACC.PL, neither PAR nor GEN may intervene between NOM and ACC, so NOM must immediately precede (<¹) ACC in this hierarchy: NOM <¹ ACC. Otherwise, if PAR or GEN intervened between NOM and ACC, this would lead to a violation of the robust *ABA generalization.⁶³

⁶²While the content of the tables is drawn from Caha 2009: 114, the order of columns and the graying of cells is slightly different.

⁶³In most general terms, “*ABA generalization refers to morphological patterns in which, given some arrangement of the relevant forms in a structured sequence, the first and third may share some property ‘A’ only if the middle member shares that property as well” (Bobaljik and Sauerland 2018: 1–2). A common application of *ABA involves root suppletion and it is best illustrated with an example from degree morphology. Given the hierarchy of degree values: positive < comparative < superlative, ABB suppletions such as *good – better – best* are expected, but an ABA pattern such as *good – better – *goodest* is not, and this prediction seems to be cross-linguistically valid (Bobaljik 2012). Here, *ABA refers to syncretism, as in Caha 2009, rather than to root suppletion (cf. Zompì 2019).

Similarly, Estonian ACC is systematically syncretic with GEN in the singular, as illustrated in (84). If PAR.SG were also always syncretic with ACC.SG=GEN.SG, as in the ‘corridor.SG’ column, then this would tell us nothing about their relative order. However, as shown in the other three columns, PAR.SG often differs from ACC.SG=GEN.SG. This precludes the order ACC < PAR < GEN, as such an order would violate the *ABA generalization. This in turn means that GEN must be adjacent to ACC. Given that NOM <[!] ACC, this translates into: NOM <[!] ACC <[!] GEN. So much is uncontroversial.

However, the last two columns of (84) additionally show that PAR may morphologically contain GEN=ACC, so at least one of the following relations must be true: ACC < PAR or GEN < PAR. But since ACC <[!] GEN, the only possibility seems to be that PAR is more marked than *both* ACC and GEN, leading to the total order in (80), summarized in (85).

(85) NOM < ACC < GEN < PAR < obliques (cf. Caha 2009: 115)

Crucially, this conclusion is unavoidable only on the assumption – made explicitly throughout Caha 2009 – that the case hierarchy is a total order. However, this assumption has been questioned in subsequent literature, e.g., in Graf 2019, Zompì 2019, and Bány 2021 (see also Harðarson 2016 and Christopoulos and Zompì 2022). For example, Bány (2021) argues for the following partially ordered case hierarchy for Spanish:

(86) NOM < ACC < {GEN $\nless$ DAT} < ... (Bány 2021, Spanish)

Moreover, cross-linguistic considerations show that the genitive is a relatively volatile case, exhibiting different relations to accusative and dative in different languages (Harðarson 2016, Zompì 2019, Starke 2017, Bány 2021). The partially ordered case hierarchy that we proposed above, repeated below as (87), is compatible with these considerations.

(87) NOM < ACC < {GEN $\nless$ PAR} < obliques (Estonian)

Note that this partial hierarchy satisfies the uncontroversial relations NOM <[!] ACC <[!] GEN argued for in Caha 2009. Moreover, it is also compatible with the observation that PAR.SG forms such as *aasta-t* and *kirik-u-t* in table (84) contain the GEN.SG=ACC.SG forms such as *aasta* and *kirik-u*. However, the explanation here is different from Caha’s (2009): this containment is conditioned by the relation ACC < PAR in (87), not by the hypothetical GEN < PAR postulated by Caha.⁶⁴

In partial summary, both case hierarchies are consistent with the Estonian syncretism and containment data discussed in Caha 2009: the partial order in (87), which is compatible with our case-stacking analysis of Estonian pseudopartitives, and Caha’s total order in (85), which is not fully compatible with this analysis. But are there independent arguments that would make it possible to choose between these two options?

In fact, there is a rather direct argument for the partially ordered Estonian case hierarchy. As often observed, Estonian oblique cases – illative, inessive, etc. – are based on the genitive, both in the singular and in the plural. This can be observed in table (73), as well as in the tables below, which extend the above tables (83)–(84) with three oblique cases: illative, allative, and translative (the others are formed fully regularly, as in (73)).

(88) Estonian cases (structural and some oblique), plural

position	‘flag.PL’	‘church.PL’	‘person.PL’
NOM	lipud	kirikud	inimesed
ACC	lipud	kirikud	inimesed
GEN	lippude	kirikute	inimeste
PAR	lippusid	kirikusid	inimesi
ILL	lippude-sse	kirikute-sse	inimeste-sse
ALL	lippude-le	kirikute-le	inimeste-le
TRL	lippude-ks	kirikute-ks	inimeste-ks

⁶⁴ Additional data discussed in Caha 2009: 130–134 may be reanalyzed the same way.

(89) Estonian cases (structural and some oblique), singular

position	‘corridor.sg’	‘person.sg’	‘year.sg’	‘church.sg’
NOM	koridor	inimene	aasta	kirik
ACC	koridor-i	inimes-e	aasta	kirik-u
GEN	koridor-i	inimes-e	aasta	kirik-u
PAR	koridor-i	inimes-t	aasta-t	kirik-u-t
ILL	koridor-i-sse	inimes-e-sse	aasta-sse	kirik-u-sse
ALL	koridor-i-le	inimes-e-le	aasta-le	kirik-u-le
TRL	koridor-i-ks	inimes-e-ks	aasta-ks	kirik-u-ks

Now, the argument against the total order “. . . < GEN < PAR < obliques” is based on the observation that the fully regular containment of GEN within oblique cases would usually have to skip the intervening PAR, violating the *ABA generalization. This is fully clear in the case of plural forms in table (88), where, for example, *lippudesse* ‘flag.ILL.PL’ contains the genitive *lippude*, and not the partitive *lippusid*, as might be expected given Caha’s (2009) total order. Similarly, in the case of singular forms, the same argument can be made on the basis of those paradigms in which GEN ≠ PAR, especially where neither is contained in the other, as in the ‘person.sg’ column.

By contrast, these data are not problematic on the partially ordered case hierarchy assumed above, as PAR does not intervene between GEN and the oblique cases. Since PAR and GEN are not comparable, the oblique cases are immediately more marked than both GEN and PAR, and they happen to be morphologically based on the former.

In fact, the frequently mentioned generalization that oblique cases are uniformly based on the genitive is only an approximation, and the full range of data provides further support for the partially ordered case hierarchy. The crucial observation is that, in brief, while oblique cases are normally based on the genitive, they are sometimes based on the partitive. This is not surprising if Estonian obliques are immediately more marked than *both* PAR and GEN, as proposed above.

There are two situations in which Estonian obliques may be based on the partitive rather than the genitive. First, in the singular, the regular “long” illative formed with *-sse* has an alternative “short” form. This short illative may have different forms, but there is a class of roots for which it is identical to the partitive, e.g.:⁶⁵

(90) Estonian singular illative (some lexemes)

position	‘bird.sg’		‘hoof.sg’		‘smith.sg’	
NOM	lind		kabi		sepp	
GEN	linnu		kabja		sepa	
PAR		lindu		kapja		seppa
ILL	linnu-sse	lindu	kabja-sse	kapja	sepa-sse	seppa
ALL	linnu-le		kabja-le		sepa-le	
TRL	linnu-ks		kabja-ks		sepa-ks	

As convincingly argued in Hirvonen 2020, short illative is a form of illative, rather than a separate case, as sometimes assumed: it occurs in the same positions as the regular illative, short and long illative forms of nouns and adjectives may co-occur within a single nominal phrase, etc. Such data are readily compatible with the view, defended above, that Estonian oblique cases are immediately more marked than *both* GEN and PAR, but less clearly compatible with the totally ordered case hierarchy, on which Estonian obliques would be immediately more marked than *only* GEN.⁶⁶

While in the singular such “parallel forms” (as they are called in Estonian linguistics) are only possible in ILL, with other obliques uniformly based on the genitive (see ALL and TRL in table (90)), in the plural some lexemes allow for similarly parallel forms of all true obliques, and when they do, these additional forms are sometimes based on PAR, as illustrated in the following table:

⁶⁵ Accusative is omitted in (90)–(91); recall that ACC.SG=GEN.SG and ACC.PL=NOM.PL.

⁶⁶ However, these data may also be analyzed in purely phonological terms, as in Hirvonen 2020, which weakens the current argument. The next argument, based on plural forms of obliques, does not seem amenable to such a purely phonological analysis.

(91) Estonian plural obliques (some lexemes)

position	'person.PL'		'corridor.PL'	
NOM	inimesed		koridorid	
GEN	inimeste		koridoride	
PAR		inimesi		koridore
ILL	inimeste-sse	inimesi-sse	koridoride-sse	koridore-sse
INE	inimeste-s	inimesi-s	koridoride-s	koridore-s
ELA	inimeste-st	inimesi-st	koridoride-st	koridore-st
ALL	inimeste-le	inimesi-le	koridoride-le	koridore-le
ADE	inimeste-l	inimesi-l	koridoride-l	koridore-l
ABL	inimeste-lt	inimesi-lt	koridoride-lt	koridore-lt
TRL	inimeste-ks	inimesi-ks	koridoride-ks	koridore-ks

Again, such effects are expected if oblique cases are immediately more marked than *both* PAR and GEN, but surprising if one of these two cases intervenes between the other and the obliques.

In summary, the partial order of cases needed to account for the heterogeneous vs. homogeneous pattern in Estonian pseudopartitives has an independent – morphological – motivation and, hence, it satisfies Norris’s (2018) objective “to avoid a stipulated case hierarchy”.

6.5 Comparison to Norris 2018

As noted above, Norris (2018) rejects an analysis based on case stacking and exponence rules. Instead, he offers an account based on the *Case Concord* principle in (92) and the *Unmarked Case* principle in (93):

(92) *Case Concord*:

Let X and Y be two nodes in a single extended projection, Y immediately dominating X.

If Y has a valued case feature [CASE: α] (but X does not), then copy Y’s case feature to X.

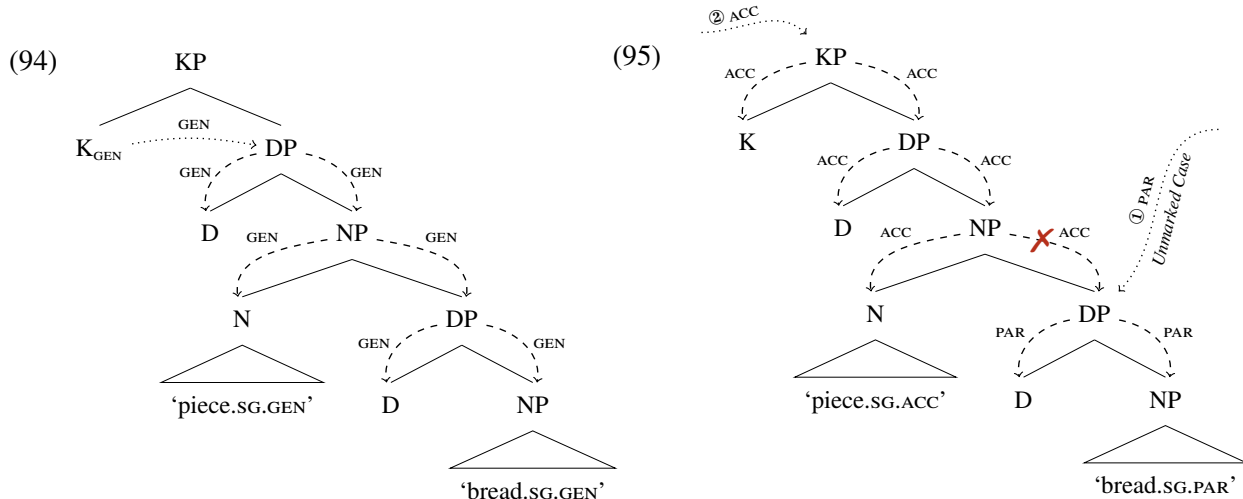
(93) *Unmarked Case* (Estonian KP):

Assign partitive case to complements of N^0 that do not already have a case value when the pseudopartitive extended projection is complete.

Case Concord ensures that a case feature on a node percolates downward. It is parameterized: it may hold in some languages but not in others, and when it does hold, case may percolate to nodes that already have case (accounting for overt case stacking in some languages) or not (accounting for languages such as Estonian). Note also that this principle only accounts for case concord, not for concord in ϕ features.

Similarly, *Unmarked Case* depends on language (here, Estonian) and maximal extended projection (here, KP). This principle is unusual in the sense that it seems to boil down to nominal heads assigning the partitive case to their complements, but unlike other such assignments, this one is context-sensitive: it happens only once the maximal extended projection of this nominal head – a KP – is constructed. For this reason it is packaged as a rule responsible for assigning the unmarked case in the sense of Dependent Case Theory (DCT; Marantz 1991, Bobaljik 2008, Baker 2015), but unlike other such rules postulated in the DCT literature, this one is limited to a specific constituent in a specific position, rather than any case-less nominal projection within a given domain.

The analysis of Estonian pseudopartitives is based on the additional assumption that when K merges with DP, it may assign a specific oblique case such as GEN or ILL, or it may be unspecified for case, in which case it receives NOM or ACC later on in the derivation via DCT principles, depending on the position of the KP. If K assigns a specific case, then this case percolates down according to the *Case Concord* principle before the *Unmarked Case* has a chance to mark the relevant DP with PAR; this is illustrated in (94) below, with dashed lines indicating case copying effects of *Case Concord*. On the other hand, when K is not marked for a specific case, then the *Unmarked Case* causes the DP to bear PAR. As illustrated in (95) below, this stops the percolation of the NOM or ACC assigned later on to the whole KP, as Estonian is parameterized for the presence of the “but X does not” bit of the *Case Concord* principle in (92).



We claim that the approach to concord advocated in previous sections is theoretically leaner. Where the approach in Norris 2018 postulates two syntactic principles, one of which is specific to case concord (with other principles needed for concord in ϕ features; cf. Norris 2014) and the other is rather ad hoc, the current approach is based on independently motivated and commonly assumed syntactic mechanisms (Agree, locality), with the one new morphological principle – *Expone Outer Most Marked* – generalizing exponence rules of stacked feature bundles proposed previously (e.g., Baker and Vinokurova 2010, Richards 2013, Pesetsky 2013, Ionin and Matushansky 2018: §6.3), also outside the context of nominal concord (e.g., Clem and Deal 2026, Grishin 2024). Moreover, as demonstrated above and in the following subsection, cross-linguistic variation on the current approach is a matter of properties of functional and semi-lexical heads, which is more in line with the Borer-Chomsky Conjecture⁶⁷ than the two parameterizable principles in (92)–(93).

Arguably, the current approach is also superior empirically. For example, as case concord is dissociated from ϕ concord in Norris 2014, 2018, it is not clear how to translate to Norris’s (2018) framework the analysis of Russian paucals in §5.2.2, where the counted noun in the genitive bears the *paucal* number in NOM and ACC positions but the *plural* number in GEN.

The following section demonstrates that the cross-linguistic data presented in Norris 2018 as “promising” for the approach to concord presented there also fall out naturally from the approach proposed in previous sections.

6.6 Finnish and Other Languages

As discussed in Norris 2018: §5.2 on the basis of data mainly from Brattico 2008, 2010, 2011, there is an interesting contrast between Estonian and the closely related Finnish: while Estonian pseudopartitives show the heterogeneous vs. homogeneous pattern discussed above, Finnish displays a strictly heterogeneous pattern, with N2 always in the partitive, even in oblique positions, e.g. (Norris 2018: 554, (65b), (64b)):

- | | |
|---|--|
| <p>(96) selle-st hulga-st auto-de-st
 this-ELA group-ELA car-PL-<u>ELA</u>
 ‘from this group of cars’ (EST)</p> | <p>(97) siitä jouko-sta auto-ja
 that.ELA set-ELA car-PL-<u>PAR</u>
 ‘from that set of cars’ (FIN)</p> |
|---|--|

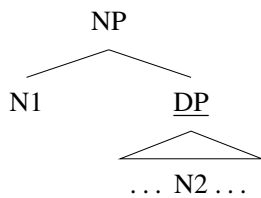
This is a little surprising, as in both languages the same heterogeneous vs. homogeneous pattern is witnessed in numeral phrases: the counted noun is in the partitive in NOM and ACC positions, and in the appropriate agreeing case in other positions.

Norris 2018: 555 analyzes the difference illustrated in (96)–(97) as follows:

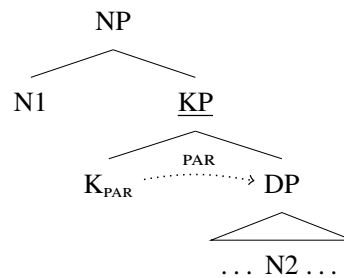
⁶⁷As formulated in Baker 2008: 156, (3):

- (i) *The Borer-Chomsky Conjecture*
All parameters of variation are attributable to differences in the features of particular items (e.g., the functional heads) in the lexicon.

(98) Estonian pseudopartitives:



(99) Finnish pseudopartitives:



First, the two languages differ in the structure of pseudopartitives: the complement of the N1 is a DP in Estonian but a KP in Finnish. Second, the partitive in Finnish pseudopartitives is inherent, unlike the unmarked partitive in Estonian pseudopartitives and in both Finnish and Estonian numeral phrases. This second assumption is needed to ensure that the “but X does not” bit of the *Case Concord* principle in (92) blocks in Finnish the downward percolation of whatever case the whole pseudopartitive construction receives.

By contrast, only the first of these two differences between Estonian and Finnish needs to be assumed on the current analysis: since the complement of N1 is a KP in Finnish, no constituent within this KP may be flagged with an external case by virtue of standard assumptions about *horizons*.

Norris (2018: §5.3) also considers some cross-linguistic variation in the marking of the N2 in pseudopartitives, suggesting that it results from the fact that languages are parameterized for the presence or absence of each of the two principles in (92)–(93), as shown in the following table:

(100) Marking of the N2 phrase in pseudopartitives (Norris 2018: 558, Table 3):

		<i>Unmarked Case</i>	
		+	–
<i>Case Concord</i>	+	Estonian	German, Greek
	–	English	Danish

Thus, Estonian has both *Case Concord* (CC) and *Unmarked Case* (UC), German and Greek have CC but not UC, English has UC (realized as the preposition *of*) but not CC, while Danish has neither:

(101) zwei Gläser Wasser (DEU; +CC, –UC)

two glasses.NOM water.NOM

(102) to glas vand (DAN; –CC, –UC)

two glasses water

‘two glasses of water’ (ENG; –CC, +UC)

On the current approach, this variation is explained equally readily, without any specialized concord principles; it is sufficient to replace $\pm$ *Case Concord* in (100) with $\pm$ *morphological case*, and $\pm$ *Unmarked Case* with $\pm$ *case flagging*. Thus, in German and Greek, which – just like Estonian – have morphological case on lexical nouns, N1 does not assign any case to its DP complement, so the case assigned by the higher K does not compete with any other case on N1 and, even when that case is the most unmarked NOM or ACC, it may be realized overtly (see (101)). By contrast, English and Danish do not have nominal morphological case, and even their pronominal case systems are occasionally analyzed as plain allomorphy (see, e.g., Hudson 1995, as well as Weisser 2020 and references therein), so no case suffixes are observed there (see (102)). The difference between English and Danish is that the complement of N1 is marked with *of* in English, but unmarked in Danish – a fact that is easy to model by appropriate properties of the English semi-lexical N1.⁶⁸

⁶⁸If *of* is analyzed as a preposition, these would be selectional properties of N1. If it is analyzed as a phrasal case marker, these would be case-flagging properties of N1, as suggested in Norris 2018: 557 and in the main text. We remain agnostic on this issue, noting only that either analysis is immediate and does not require specialized concord principles.

6.7 Interim Summary and Comparisons

In summary, Finnic data confirm the analysis of concord as insatiable Agree proposed in previous sections. The difference between Estonian and Finnish may be analyzed as in Norris 2018, in terms of different selectional properties of N1, but unlike in the analysis of Norris 2018, no additional assumptions about the different status of the partitive in Estonian and Finnish are necessary. Moreover, the heterogeneous vs. homogeneous pattern in Estonian pseudopartitives is explained by exactly the same case stacking and exponence rules as similar patterns in Slavic numerals, once the partial order of the Estonian case hierarchy is established on independent morphological grounds. The fact that the *Expone Outer Most Marked* rule proposed on the basis of Russian paucals is also needed to account for Estonian pseudopartitives suggests that this approach to exponence may be cross-linguistically valid.

This last conclusion is supported by the fact that this rule seems to have the desired effect in many analyses where other exponence rules have been proposed. First, it accounts for the Russian facts for which the *Expone Outermost* was invoked (Pesetsky 2013). Second, it gives the right result where the rule *Expone Most Marked* was used, again including Russian numerals (Matushansky 2008, Ionin and Matushansky 2018), but also Gothic free relatives (Grishin 2024). Finally, in their analysis of Sakha, Baker and Vinokurova (2010) invoke a rule that could be called *Expone Innermost*, in order to account for the dative case on nominals first assigned DAT and then ACC, i.e., with the DAT.ACC stack. Given that nominals with the NOM.ACC stack expone the outermost accusative, they need to add a special proviso to this rule, to the effect that morphologically unmarked cases, such as NOM, do not count when determining the innermost case. By contrast, both the *Expone Most Marked* rule and our *Expone Outer Most Marked* variant predict that DAT.ACC is expounded as DAT and NOM.ACC as ACC, given the commonly assumed markedness order $\text{NOM} < \text{ACC} < \text{DAT}$ (Blake 2001, Caha 2009, Smith *et al.* 2019, Zompì 2019, 2023).

7 Conclusion

The main theoretical contribution of this paper is a demonstration that, given the interaction/satisfaction model of Agree and the *horizons* approach to locality, both extensively motivated by phenomena independent of nominal concord, no specialized feature percolation mechanisms are needed to account for concord, contrary to the currently dominant view. The approach to concord proposed here has been torture tested by applying it to the most difficult aspects of nominal concord we are aware of, namely, the mixed homogeneous vs. heterogeneous case and ϕ patterns in Finnic and Slavic, including perhaps the most mind-boggling such pattern, witnessed in Russian paucal numeral constructions.

These tests did not break the proposed approach to concord but they made clear the need to refine a number of earlier theoretical proposals. First, while the phenomena considered in Keine 2019, 2020 do not determine whether only constituents properly contained within *horizons* are inaccessible to probes, or whether the *horizons* themselves are also off-limits, concord clearly points to the latter option. Second, while various principles have been proposed regulating exponence of items with multiple feature bundles, a single rule generalizing *Expone Outermost* and *Expone Most Marked* turned out to be needed to account for the most complex Slavic and Finnic patterns, as well as sufficient for many patterns previously handled by those earlier rules; however, it remains to be verified whether this single rule is sufficient for all phenomena of this kind, or whether different variants of such exponence rules are needed for different phenomena, as suggested, e.g., in Grishin 2024. Third, in the process of applying this approach to Finnic, we had to refine the case markedness hierarchy for Estonian; in particular, we demonstrated that it is not totally ordered, adding an argument to the growing literature on non-linear case hierarchies. Finally, while most of the work in nominal concord is carried out by insatiable probes, we confirmed the need for the satiable/insatiable distinction, as it would be difficult to account for the feminine pattern of Russian paucals without it.

Relatedly, the main empirical contribution of the current paper is an account of Russian paucals. Because of their complexity and instability, previous generative accounts often concentrated on particular aspects of their morphosyntax, ignoring others. We believe that only a comprehensive treatment, such as the one offered above, taking into account not only the celebrated differences in stress patterns in 5 masculine nouns, but also the form of de-adjectival nouns, the dependence of adjectival forms on gender, the fact that there are fractional paucals, etc., makes it possible to formulate a robust analysis. In developing such an analysis, we argued for the “paucal number”

hypothesis, even though the “paucal case” analysis is easier and more immediately compatible with the approach to concord proposed here. To this end, we summarized known arguments for “paucal number” (e.g., the argument from the violation of the “no gender distinctions in the plural” generalization in Russian), and adduced new ones (esp., from violations of *ABA and certain cross-linguistic generalizations about stem allomorphy).

We can only hope that these theoretical and empirical contributions of this paper will inspire future work.

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